# TM Forum Specification

# Alarm

**TMF642**

| | |
|---|---|
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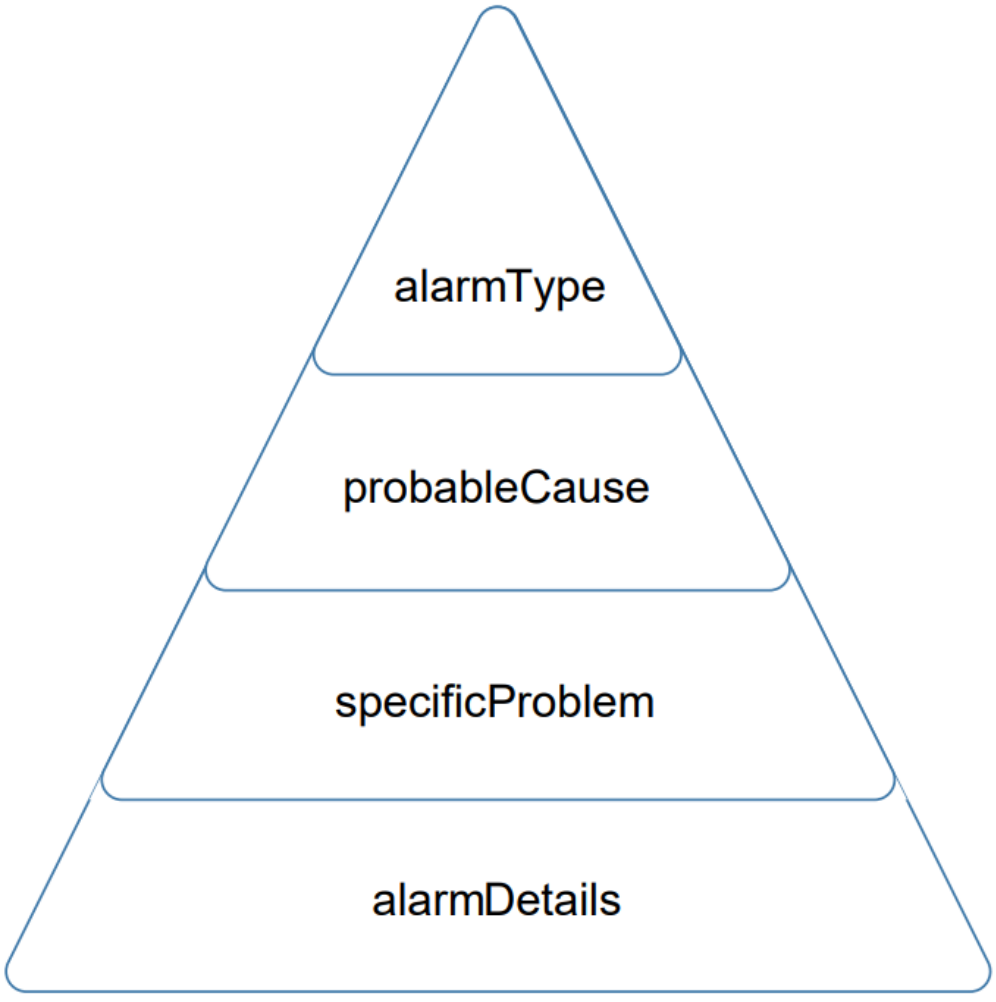
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# Introduction

The following document is the user guide of the REST API for Alarm management. It includes the model definition as well as all available operations.

## Alarm Information

An alarm is described in increasing detail by the attributes shown in the diagram below. Every alarm must include an **alarmType** which places the fault within a small set of broad categories. A **probableCause** is also required and attributes one of an industry agreed set of descriptions of the probable cause. Vendor specific descriptions may be contained in **specificProblem** and more detailed information about the fault may be provided in **alarmDetails**.




## Alarm Type

The categories for alarm classification are defined in X.733 8.1.1 and 3GPP TS 32.111-2 Annex A. The complete enumerated list of possible values for the mandatory alarmType attribute is as follows:

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| alarmType | Description |
|---|---|
| communicationsAlarm | A communications fault. |
| processingErrorAlarm | A software or processing fault. |
| environmentalAlarm | A condition related to an enclosure in which the equipment resides. |
| qualityOfServiceAlarm | A degradation in the quality of a service. |
| equipmentAlarm | An equipment fault. |
| integrityViolation | Information may have been illegally modified, inserted or deleted. |
| operationalViolation | The unavailability, malfunction or incorrect invocation of a service. |
| physicalViolation | A physical resource has been violated in a way that suggests a security attack. |
| securityService | A security attack has been detected by a security service. |
| mechanismViolation | A security attack has been detected by a security mechanism. |
| timeDomainViolation | An event has occurred at an unexpected or prohibited time. |

## Probable Cause

Probable Cause values are defined in ITU-T X.733 and 3GPP TS 32.111-2 Annex B. It is highly recommended to choose a value from the existing list. If none of the values from standarts are satisfactory consider choosing "other" and describe your situation with the specificProblem attribute.

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# Sample Use Cases

The Alarm Management API provides the standardized client interface to Alarm Management systems for creating, tracking and managing alarms among partners. The interface supports alarm management on both resources and services. The alarmed objects are not restricted to any particular technology or a vendor, so the API can be used in a wide variety of fault management cases.

In real-life deployments we see various levels of fault management API needs starting from simple subscription on alarm lifecycle events, up to full synchronization of acknowledgements and root cause analysis between two alarm management systems.

Two main kinds of business scenarios were identified:

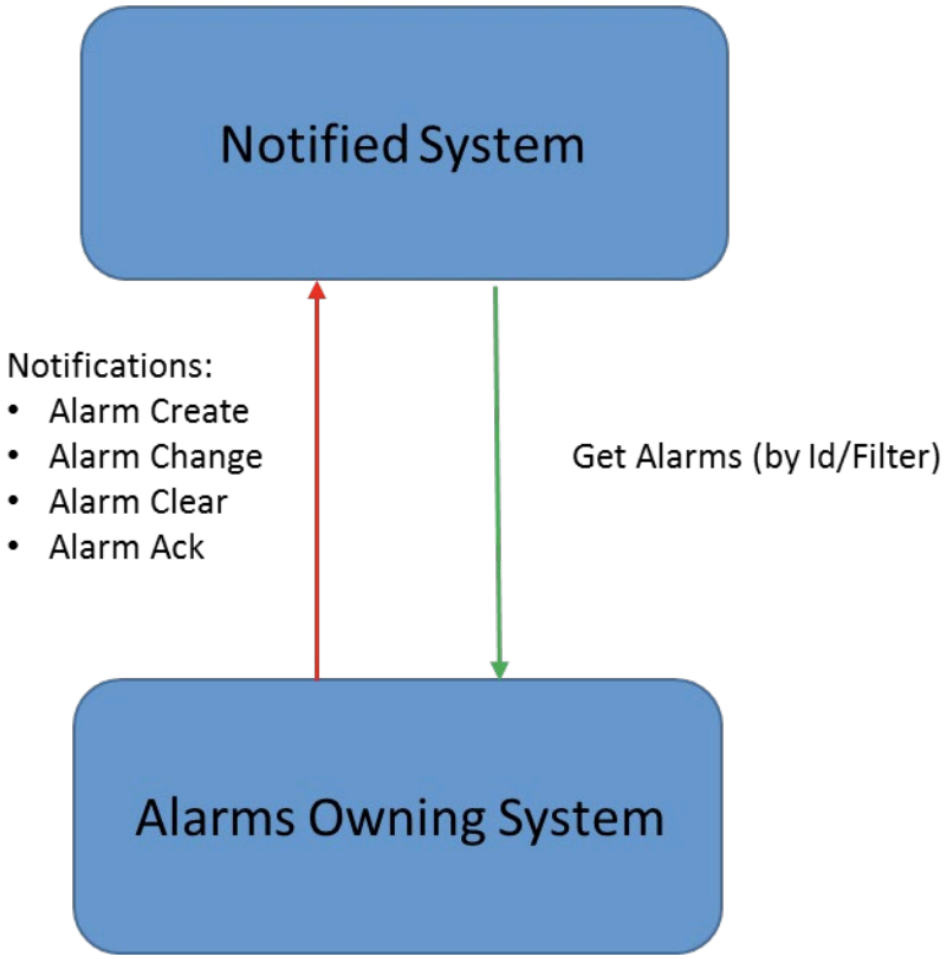
- Management Functions subscriptions ("simple" alarm notifications)
- Synchronization of Management systems (on alarm events, threshold crossing alarms, acknowledgements, root cause analysis, etc.)

## Sample Use Case – Simple Alarm Forwarding

In the first case, one party of the interface is an Alarm Management system (the alarm-owning system) and the second party is a management function that is subscribed on events, mainly alarm life-cycle events. It cannot be assumed that the subscribed function has a persistent view of the alarms, as it is not necessarily an alarm management system. The subscriber party can be a UI, a communication hub, a Service Quality management system, a BSS system, or any other function that is interested in alarm events. In this case, the operations that will be used are typically:

- Alarm life-cycle notifications:
  - Raise notification (mandatory)
  - Clear notification(mandatory)
  - Change notification (optional)
- Get Alarms operations: used by the Management Function to get synchronized on the state of active alarms in situations where snapshots of the active alarms are required, such as system start, or recovery from communication failures. This operation may include a filter on the subset of alarms that are of interest.
- The acknowledgement notification (optional)

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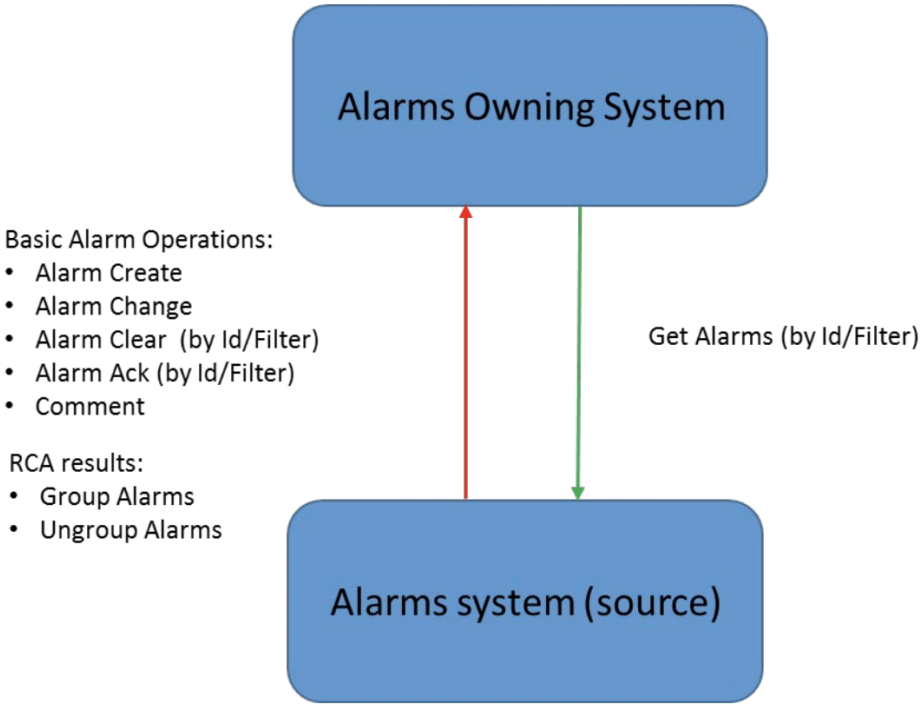

## Sample Use Case – OSS-to-OSS

The second case is where the two parties are both alarm management systems/functions and they have to synchronize alarms in different aspects. Typically, Alarm Management system A is one of the alarm data sources of Alarm management system Z. In this case the operations will be slightly different with a tighter integration:

- Alarm management system A can raise, change and clear alarms in Alarm Management system Z
- Alarm management system A can acknowledge alarms in Alarm Management system Z
- Alarm Management system A can apply root cause analysis results in Alarm Management system Z by using the Group and Ungroup operations.
- Alarm management system A can comment (annotate) alarms in Alarm Management system Z
- Get Alarms operations used by the Management Function to get synchronized on the state of active alarms in situations where snapshots of the active alarms are required, such as system start, or recovery from communication failures. This operation may include a filter on the subset of alarms that are of interest.

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In this scenario, since the level of integration is tighter, it is important that Alarm Management System A gets the information on the success of the operations in Alarm management system Z.




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# Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a BillingAccount and SettlementAccount inheriting properties from the abstract Account entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Account instances some may be instances of BillingAccount where other could be instances of SettlementAccount. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an AccountRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as BillingAccountRef or SettlementAccountRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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# RESOURCE MODEL

## Managed Entity and Task Resource Models

### Alarm resource

This resource represents an alarm supporting the information model defined in ITU-T X.733.

**Resource model**

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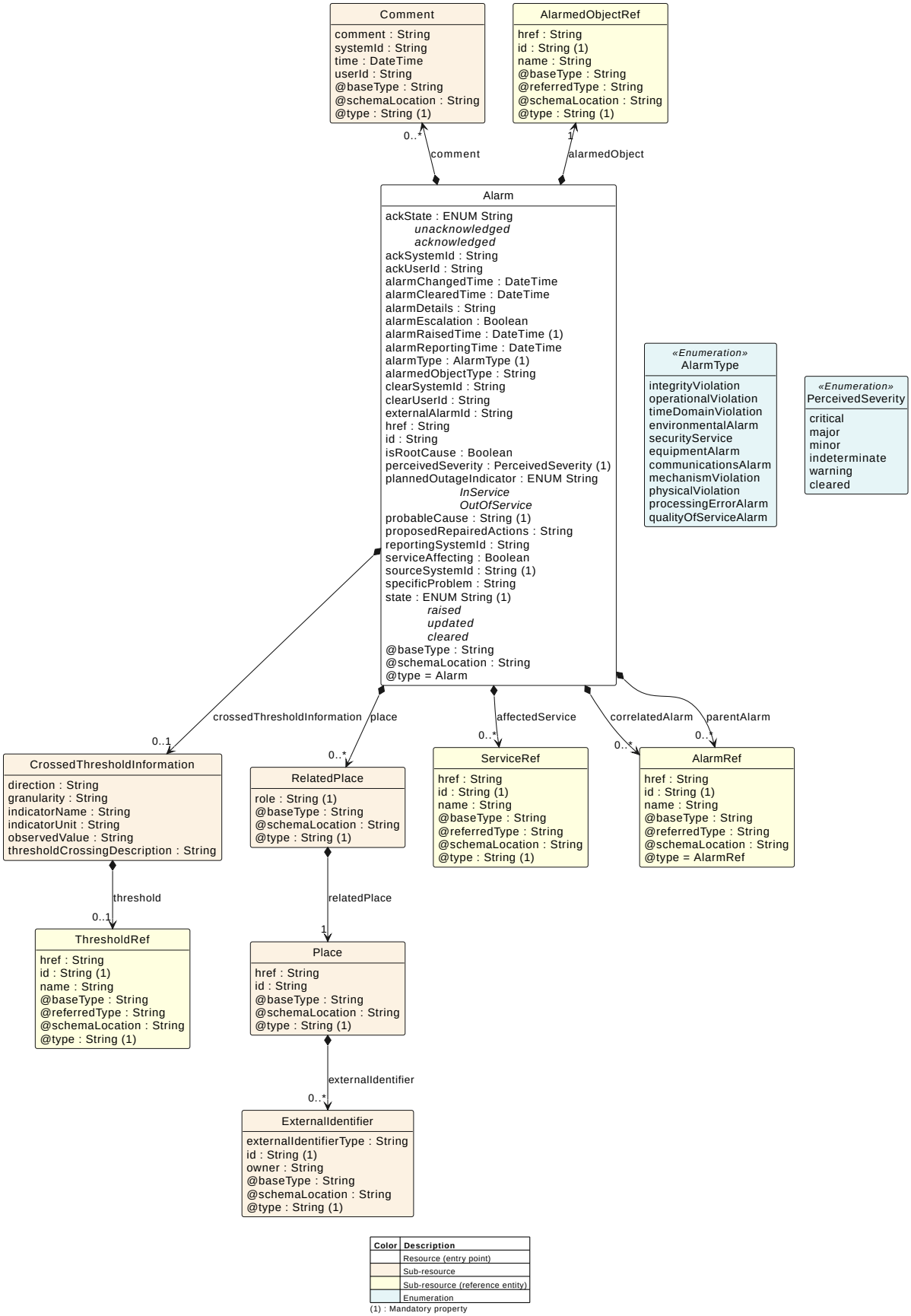



*Figure 1 - Alarm*

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*Figure 2 - Alarm*

## Field descriptions

### Alarm fields

| | |
|---|---|
| ackState | A String. Provides the Acknowledgement State of the alarm (unacknowledged, acknowledged). ENUMERATED with values:<br>* unacknowledged<br>* acknowledged |
| ackSystemId | A String. Provides the name of the system that last changed the ackState of an alarm, i.e. acknowledged or unacknowledged the alarm. |
| ackUserId | A String. Provides the id of the user who has last changed the ack state of the alarm, i.e. acknowledged or unacknowledged the alarm. |
| affectedService | A ServiceRef. Service reference, for when Service is used by other entities. |
| alarmChangedTime | A DateTime. Indicates the last date and time when the alarm is changed on the alarm-owning system. Any change to the alarm whether coming from the alarmed resource, or triggered by a change from the client is changing this time. |
| alarmClearedTime | A DateTime. Indicates the time (as a date + time) at which the alarm is cleared at the source. |
| alarmDetails | A String. Contains further information on the alarm. |
| alarmEscalation | A Boolean. Indicates if this alarm has been escalated or not. |
| alarmRaisedTime | A DateTime. Indicates the time (as a date + time) at which the alarm occurred at its source. |
| alarmReportingTime | A DateTime. Indicates the time (as a date + time) at which the alarm was reported by the owning OSS. It might be different from the alarmRaisedTime. For instance, if the alarm list is maintained by an EMS, the alarmRaisedtime would be the time the alarm was detected by the NE, while the alarmReportingTime would be the time this alarm was stored in the alarm list of the EMS. |
| alarmType | An AlarmType. Categorizes the alarm (X.733 8.1.1, 3GPP TS 32.111-2 Annex A). ENUMERATED with values:<br>* communicationsAlarm<br>* processingErrorAlarm<br>* environmentalAlarm<br>* qualityOfServiceAlarm<br>* equipmentAlarm<br>* integrityViolation<br>* operationalViolation<br>* physicalViolation<br>* securityService<br>* mechanismViolation<br>* timeDomainViolation |

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| | |
|---|---|
| alarmedObject | An AlarmedObjectRef. Reference to object which affected by Alarm (AlarmedObject). |
| alarmedObjectType | A String. The type (class) of the managed object associated with the event. |
| clearSystemId | A String. Provides the id of the system where the user who invoked the alarmCleared operation is located. |
| clearUserId | A String. Provides the id of the user who invoked the alarmCleared operation. |
| comment | A Comment. Indicates the comments entered on the alarm. |
| correlatedAlarm | An AlarmRef. |
| crossedThresholdInformation | A CrossedThresholdInformation. Identifies the details of the threshold that has been crossed. |
| externalAlarmId | A String. An identifier of the alarm in the source system. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| isRootCause | A Boolean. Indicates whether the alarm is a root cause alarm.. |
| parentAlarm | An AlarmRef. |
| perceivedSeverity | A PerceivedSeverity. Lists the possible severities that can be allocated to an Alarm. The values are consistent with ITU-T Recommendation X.733. Once an alarm has been cleared, its perceived severity is set to 'cleared' and can no longer be set. ENUMERATED with values:<br>* critical<br>* major<br>* minor<br>* warning<br>* indeterminate<br>* cleared |
| place | A RelatedPlace. A Place and an associated role as installation address, delivery address, etc.… |
| plannedOutageIndicator | A String. Indicates that the Managed Object (related to this alarm) is in planned outage (in planned maintenance, or out-of-service). ENUMERATED with values:<br>* InService<br>* OutOfService |
| probableCause | A String. Provides the probable cause of the alarm. The values are consistent with ITU-T Recommendation X.733 or 3GPP TS 32.111-2 Annex B. |
| proposedRepairedActions | A String. Indicates proposed repair actions, if known to the system emitting the alarm. |
| reportingSystemId | A String. Reporting system identity. |
| serviceAffecting | A Boolean. Indicates whether the alarm affects service or not. |
| sourceSystemId | A String. Source system identity. |

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| | |
|---|---|
| specificProblem | A String. Provides more specific information about the alarm. |
| state | A String. Defines the alarm state during its life cycle (raised, updated, cleared). ENUMERATED with values:<br>* raised<br>* updated<br>* cleared |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Comment sub-resource fields

| | |
|---|---|
| comment | A String. Indicates the text of the comment. |
| systemId | A String. Indicates the system identifier on which the client set the comment. |
| time | A DateTime. Indicates the time commenting the alarm. |
| userId | A String. Indicates the user commenting the alarm. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### AlarmedObjectRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### CrossedThresholdInformation sub-resource fields

| | |
|---|---|
| direction | A String. Indicates the threshold crossing direction: up or down. |
| granularity | A String. Indicates the granularity at which the indicator is evaluated for threshold crossing. |

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| | |
|---|---|
| indicatorName | A String. Indicates the name of indicator which crossed the threshold. |
| indicatorUnit | A String. Indicates the unit of the measurement of the indicator corresponding to the threshold that has been crossed. |
| observedValue | A String. Indicates the value of the indicator which crossed the threshold. |
| threshold | A ThresholdRef. |
| thresholdCrossingDescription | A String. Indicates further information on the threshold crossing alarm. |

### RelatedPlace sub-resource fields

| | |
|---|---|
| relatedPlace | A Place. Place reference. |
| role | A String. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ServiceRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### AlarmRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |

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| | |
|---|---|
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ThresholdRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Place sub-resource fields

| | |
|---|---|
| externalIdentifier | An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ExternalIdentifier sub-resource fields

| | |
|---|---|
| externalIdentifierType | A String. Type of the identification, typically would be the type of the entity within the external system. |
| id | A String. Identification of the entity within the external system. |
| owner | A String. Name of the external system that owns the entity. |

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| | |
|---|---|
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Json representation sample(s)

We provide below a JSON representation as example of the 'Alarm' resource object.


```
{
    "id": "8675309",
    "href": "/alarmManagement/v5/alarm/8675309",
    "@type": "Alarm",
    "externalAlarmId": "5551212",
    "state": "updated",
    "alarmType": "environmentalAlarm",
    "perceivedSeverity": "major",
    "probableCause": "rectifierLowVoltage",
    "specificProblem": "ps=3,sl=1,in=8",
    "alarmedObjectType": "Rectifier",
    "alarmedObject": {
        "@type": "AlarmedObjectRef",
        "@referredType": "Resource",
        "id": "93051825",
        "href": "/resourceInventoryManagement/v5/resource/93051825"
    },
    "sourceSystemId": "ems-1",
    "alarmDetails": "voltage=95",
    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
    "alarmChangedTime": "2019-07-03T03:32:52.744Z",
    "ackSystemId": "ems-1",
    "ackUserId": "bob@example.net",
    "ackState": "acknowledged",
    "parentAlarm": [
        {
            "@type": "AlarmRef",
            "id": "8675300"
        }
    ],
    "correlatedAlarm": [
        {
            "@type": "AlarmRef",
            "id": "8675399",
            "href": "/alarmManagement/v5/alarm/868675399"
        }
    ],
    "comment": [
        {
            "@type": "Comment",
            "userId": "bob@example.net",
            "systemId": "ems-1",
            "time": "2019-07-03T03:37:33.827Z",
            "comment": "Dispatched"
        }
    ]
```


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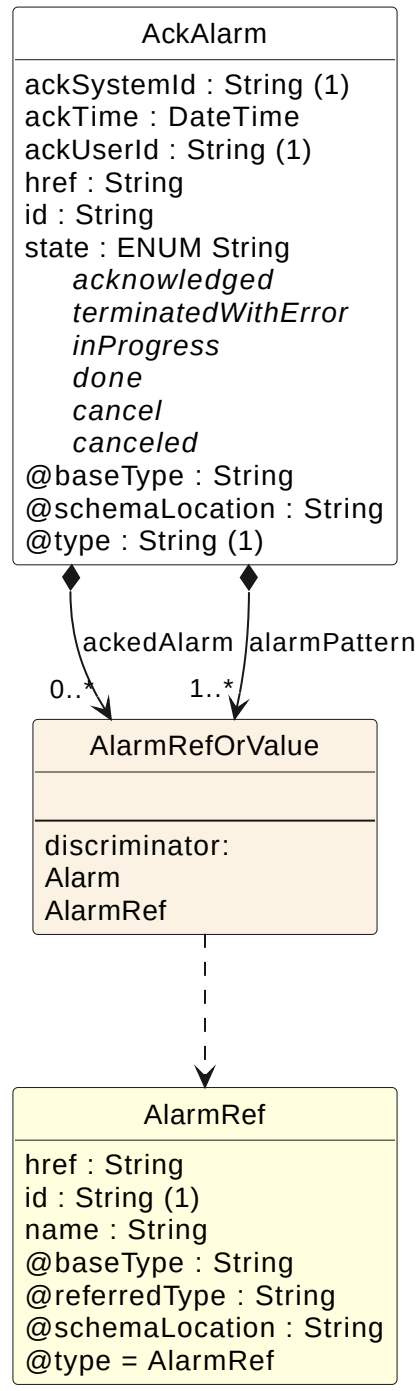
```
}
```

## AckAlarm resource

Task resource for the acknowledge alarms operation.

### Resource model

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| Color | Description |
|---|---|
| | Resource (entry point) |
| | Sub-resource |
| | Sub-resource (reference entity) |

(1) : Mandatory property

*Figure 3 - AckAlarm*

[.] | .//

*Figure 4 - AckAlarm*

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## Field descriptions

### AckAlarm fields

| | |
|---|---|
| ackSystemId | A String. Name of the acknowledging system. |
| ackTime | A DateTime. Time of the acknowledgement. |
| ackUserId | A String. Name of the acknowledging user. |
| ackedAlarm | An AlarmRefOrValue. An alarm defined by reference or value. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Alarm entity and not the AlarmRefOrValue class itself. |
| alarmPattern | An AlarmRefOrValue. An alarm defined by reference or value. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Alarm entity and not the AlarmRefOrValue class itself. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| state | A String. Current state of the operation task. ENUMERATED with values:<br>* acknowledged<br>* terminatedWithError<br>* inProgress<br>* done<br>* cancel<br>* canceled |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Alarm sub-resource fields

| | |
|---|---|
| ackState | A String. Provides the Acknowledgement State of the alarm (unacknowledged, acknowledged). ENUMERATED with values:<br>* unacknowledged<br>* acknowledged |
| ackSystemId | A String. Provides the name of the system that last changed the ackState of an alarm, i.e. acknowledged or unacknowledged the alarm. |
| ackUserId | A String. Provides the id of the user who has last changed the ack state of the alarm, i.e. acknowledged or unacknowledged the alarm. |
| affectedService | A ServiceRef. Service reference, for when Service is used by other entities. |

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| | |
|---|---|
| alarmChangedTime | A DateTime. Indicates the last date and time when the alarm is changed on the alarm-owning system. Any change to the alarm whether coming from the alarmed resource, or triggered by a change from the client is changing this time. |
| alarmClearedTime | A DateTime. Indicates the time (as a date + time) at which the alarm is cleared at the source. |
| alarmDetails | A String. Contains further information on the alarm. |
| alarmEscalation | A Boolean. Indicates if this alarm has been escalated or not. |
| alarmRaisedTime | A DateTime. Indicates the time (as a date + time) at which the alarm occurred at its source. |
| alarmReportingTime | A DateTime. Indicates the time (as a date + time) at which the alarm was reported by the owning OSS. It might be different from the alarmRaisedTime. For instance, if the alarm list is maintained by an EMS, the alarmRaisedtime would be the time the alarm was detected by the NE, while the alarmReportingTime would be the time this alarm was stored in the alarm list of the EMS. |
| alarmType | An AlarmType. Categorizes the alarm (X.733 8.1.1, 3GPP TS 32.111-2 Annex A). ENUMERATED with values:<br>* communicationsAlarm<br>* processingErrorAlarm<br>* environmentalAlarm<br>* qualityOfServiceAlarm<br>* equipmentAlarm<br>* integrityViolation<br>* operationalViolation<br>* physicalViolation<br>* securityService<br>* mechanismViolation<br>* timeDomainViolation |
| alarmedObject | An AlarmedObjectRef. Reference to object which affected by Alarm (AlarmedObject). |
| alarmedObjectType | A String. The type (class) of the managed object associated with the event. |
| clearSystemId | A String. Provides the id of the system where the user who invoked the alarmCleared operation is located. |
| clearUserId | A String. Provides the id of the user who invoked the alarmCleared operation. |
| comment | A Comment. Indicates the comments entered on the alarm. |
| correlatedAlarm | An AlarmRef. |
| crossedThresholdInformation | A CrossedThresholdInformation. Identifies the details of the threshold that has been crossed. |
| externalAlarmId | A String. An identifier of the alarm in the source system. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| isRootCause | A Boolean. Indicates whether the alarm is a root cause alarm.. |

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| | |
|---|---|
| parentAlarm | An AlarmRef. |
| perceivedSeverity | A PerceivedSeverity. Lists the possible severities that can be allocated to an Alarm. The values are consistent with ITU-T Recommendation X.733. Once an alarm has been cleared, its perceived severity is set to 'cleared' and can no longer be set. ENUMERATED with values:<br>* critical<br>* major<br>* minor<br>* warning<br>* indeterminate<br>* cleared |
| place | A RelatedPlace. A Place and an associated role as installation address, delivery address, etc.… |
| plannedOutageIndicator | A String. Indicates that the Managed Object (related to this alarm) is in planned outage (in planned maintenance, or out-of-service). ENUMERATED with values:<br>* InService<br>* OutOfService |
| probableCause | A String. Provides the probable cause of the alarm. The values are consistent with ITU-T Recommendation X.733 or 3GPP TS 32.111-2 Annex B. |
| proposedRepairedActions | A String. Indicates proposed repair actions, if known to the system emitting the alarm. |
| reportingSystemId | A String. Reporting system identity. |
| serviceAffecting | A Boolean. Indicates whether the alarm affects service or not. |
| sourceSystemId | A String. Source system identity. |
| specificProblem | A String. Provides more specific information about the alarm. |
| state | A String. Defines the alarm state during its life cycle (raised, updated, cleared). ENUMERATED with values:<br>* raised<br>* updated<br>* cleared |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### AlarmRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |

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| | |
|---|---|
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Comment sub-resource fields

| | |
|---|---|
| comment | A String. Indicates the text of the comment. |
| systemId | A String. Indicates the system identifier on which the client set the comment. |
| time | A DateTime. Indicates the time commenting the alarm. |
| userId | A String. Indicates the user commenting the alarm. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### AlarmedObjectRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### CrossedThresholdInformation sub-resource fields

| | |
|---|---|
| direction | A String. Indicates the threshold crossing direction: up or down. |
| granularity | A String. Indicates the granularity at which the indicator is evaluated for threshold crossing. |
| indicatorName | A String. Indicates the name of indicator which crossed the threshold. |
| indicatorUnit | A String. Indicates the unit of the measurement of the indicator corresponding to the threshold that has been crossed. |

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| | |
|---|---|
| observedValue | A String. Indicates the value of the indicator which crossed the threshold. |
| threshold | A ThresholdRef. |
| thresholdCrossingDescription | A String. Indicates further information on the threshold crossing alarm. |

### RelatedPlace sub-resource fields

| | |
|---|---|
| relatedPlace | A Place. Place reference. |
| role | A String. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ServiceRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ThresholdRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Place sub-resource fields

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| | |
|---|---|
| externalIdentifier | An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ExternalIdentifier sub-resource fields

| | |
|---|---|
| externalIdentifierType | A String. Type of the identification, typically would be the type of the entity within the external system. |
| id | A String. Identification of the entity within the external system. |
| owner | A String. Name of the external system that owns the entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Json representation sample(s)

We provide below a JSON representation as example of the 'AckAlarm' resource object.


```
{
    "id": "1038",
    "@type": "AckAlarm",
    "href": "/alarmManagement/v5/ackAlarm/1038",
    "ackSystemId": "ems-1",
    "ackUserId": "bob@example.net",
    "ackTime": "2019-07-03T03:33:12.623Z",
    "alarmPattern": [
        {
            "@type": "AlarmRef",
            "id": "8675309"
        },
        {
            "@type": "AlarmRef",
```


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```
            "id": "8675399"
        }
    ],
    "state": "done",
    "ackedAlarm": [
        {
            "id": "8675309",
            "href": "/alarmManagement/v5/alarm/8675309",
            "@type": "Alarm",
            "externalAlarmId": "5551212",
            "state": "updated",
            "alarmType": "environmentalAlarm",
            "perceivedSeverity": "major",
            "probableCause": "rectifierLowVoltage",
            "specificProblem": "ps=3,sl=1,in=8",
            "alarmedObjectType": "Rectifier",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "93051825",
                "href": "/resourceInventoryManagement/v5/resource/93051825"
            },
            "sourceSystemId": "ems-1",
            "alarmDetails": "voltage=95",
            "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
            "alarmReportingTime": "2019-07-03T03:32:17.552Z",
            "alarmChangedTime": "2019-07-03T03:32:52.744Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackState": "acknowledged",
            "correlatedAlarm": [
                {
                    "@type": "AlarmRef",
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/868675399"
                }
            ]
        },
        {
            "id": "8675399",
            "href": "/alarmManagement/v5/alarm/8675399",
            "@type": "Alarm",
            "externalAlarmId": "5551234",
            "state": "raised",
            "alarmType": "equipmentAlarm",
            "perceivedSeverity": "minor",
            "probableCause": "powerSupplyFailure",
            "specificProblem": "ch=1,ps=3",
            "alarmedObjectType": "Chassis",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "95830731",
                "href": "/resourceInventoryManagement/v5/resource/95830731"
            },
            "sourceSystemId": "ems-1",
            "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
            "alarmReportingTime": "2019-07-03T03:32:18.102Z",
            "alarmChangedTime": "2019-07-03T03:32:52.744Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackState": "acknowledged",
            "parentAlarm": [
```


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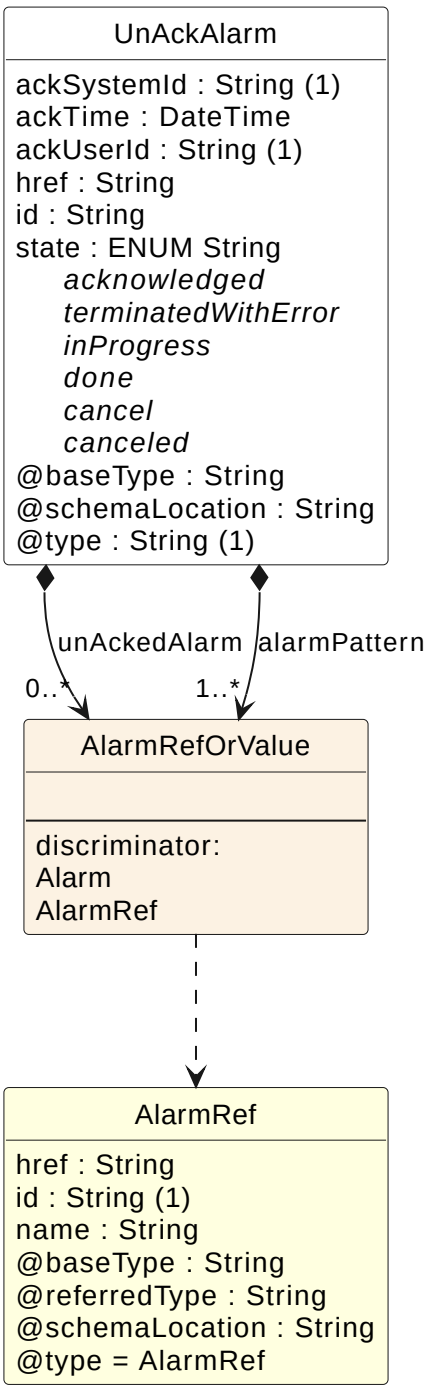
```
                {
                    "@type": "AlarmRef",
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/868675309"
                }
            ]
        }
    ]
}
```


## UnAckAlarm resource

Task resource for unacknowledge alarms operation.

### Resource model

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| Color | Description |
|---|---|
| | Resource (entry point) |
| | Sub-resource |
| | Sub-resource (reference entity) |

(1) : Mandatory property

*Figure 5 - UnAckAlarm*

[.] | .//

*Figure 6 - UnAckAlarm*

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## Field descriptions

### UnAckAlarm fields

| | |
|---|---|
| ackSystemId | A String. Name of the unacknowledging system. |
| ackTime | A DateTime. Time of the unacknowledgement. |
| ackUserId | A String. Name of the unacknowledging user. |
| alarmPattern | An AlarmRefOrValue. An alarm defined by reference or value. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Alarm entity and not the AlarmRefOrValue class itself. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| state | A String. Current state of the operation task. ENUMERATED with values:<br>* acknowledged<br>* terminatedWithError<br>* inProgress<br>* done<br>* cancel<br>* canceled |
| unAckedAlarm | An AlarmRefOrValue. An alarm defined by reference or value. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Alarm entity and not the AlarmRefOrValue class itself. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Alarm sub-resource fields

| | |
|---|---|
| ackState | A String. Provides the Acknowledgement State of the alarm (unacknowledged, acknowledged). ENUMERATED with values:<br>* unacknowledged<br>* acknowledged |
| ackSystemId | A String. Provides the name of the system that last changed the ackState of an alarm, i.e. acknowledged or unacknowledged the alarm. |
| ackUserId | A String. Provides the id of the user who has last changed the ack state of the alarm, i.e. acknowledged or unacknowledged the alarm. |
| affectedService | A ServiceRef. Service reference, for when Service is used by other entities. |

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| | |
|---|---|
| alarmChangedTime | A DateTime. Indicates the last date and time when the alarm is changed on the alarm-owning system. Any change to the alarm whether coming from the alarmed resource, or triggered by a change from the client is changing this time. |
| alarmClearedTime | A DateTime. Indicates the time (as a date + time) at which the alarm is cleared at the source. |
| alarmDetails | A String. Contains further information on the alarm. |
| alarmEscalation | A Boolean. Indicates if this alarm has been escalated or not. |
| alarmRaisedTime | A DateTime. Indicates the time (as a date + time) at which the alarm occurred at its source. |
| alarmReportingTime | A DateTime. Indicates the time (as a date + time) at which the alarm was reported by the owning OSS. It might be different from the alarmRaisedTime. For instance, if the alarm list is maintained by an EMS, the alarmRaisedtime would be the time the alarm was detected by the NE, while the alarmReportingTime would be the time this alarm was stored in the alarm list of the EMS. |
| alarmType | An AlarmType. Categorizes the alarm (X.733 8.1.1, 3GPP TS 32.111-2 Annex A). ENUMERATED with values:<br>* communicationsAlarm<br>* processingErrorAlarm<br>* environmentalAlarm<br>* qualityOfServiceAlarm<br>* equipmentAlarm<br>* integrityViolation<br>* operationalViolation<br>* physicalViolation<br>* securityService<br>* mechanismViolation<br>* timeDomainViolation |
| alarmedObject | An AlarmedObjectRef. Reference to object which affected by Alarm (AlarmedObject). |
| alarmedObjectType | A String. The type (class) of the managed object associated with the event. |
| clearSystemId | A String. Provides the id of the system where the user who invoked the alarmCleared operation is located. |
| clearUserId | A String. Provides the id of the user who invoked the alarmCleared operation. |
| comment | A Comment. Indicates the comments entered on the alarm. |
| correlatedAlarm | An AlarmRef. |
| crossedThresholdInformation | A CrossedThresholdInformation. Identifies the details of the threshold that has been crossed. |
| externalAlarmId | A String. An identifier of the alarm in the source system. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| isRootCause | A Boolean. Indicates whether the alarm is a root cause alarm.. |

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| | |
|---|---|
| parentAlarm | An AlarmRef. |
| perceivedSeverity | A PerceivedSeverity. Lists the possible severities that can be allocated to an Alarm. The values are consistent with ITU-T Recommendation X.733. Once an alarm has been cleared, its perceived severity is set to 'cleared' and can no longer be set. ENUMERATED with values:<br>* critical<br>* major<br>* minor<br>* warning<br>* indeterminate<br>* cleared |
| place | A RelatedPlace. A Place and an associated role as installation address, delivery address, etc.… |
| plannedOutageIndicator | A String. Indicates that the Managed Object (related to this alarm) is in planned outage (in planned maintenance, or out-of-service). ENUMERATED with values:<br>* InService<br>* OutOfService |
| probableCause | A String. Provides the probable cause of the alarm. The values are consistent with ITU-T Recommendation X.733 or 3GPP TS 32.111-2 Annex B. |
| proposedRepairedActions | A String. Indicates proposed repair actions, if known to the system emitting the alarm. |
| reportingSystemId | A String. Reporting system identity. |
| serviceAffecting | A Boolean. Indicates whether the alarm affects service or not. |
| sourceSystemId | A String. Source system identity. |
| specificProblem | A String. Provides more specific information about the alarm. |
| state | A String. Defines the alarm state during its life cycle (raised, updated, cleared). ENUMERATED with values:<br>* raised<br>* updated<br>* cleared |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### AlarmRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |

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| | |
|---|---|
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Comment sub-resource fields

| | |
|---|---|
| comment | A String. Indicates the text of the comment. |
| systemId | A String. Indicates the system identifier on which the client set the comment. |
| time | A DateTime. Indicates the time commenting the alarm. |
| userId | A String. Indicates the user commenting the alarm. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### AlarmedObjectRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### CrossedThresholdInformation sub-resource fields

| | |
|---|---|
| direction | A String. Indicates the threshold crossing direction: up or down. |
| granularity | A String. Indicates the granularity at which the indicator is evaluated for threshold crossing. |
| indicatorName | A String. Indicates the name of indicator which crossed the threshold. |
| indicatorUnit | A String. Indicates the unit of the measurement of the indicator corresponding to the threshold that has been crossed. |

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| | |
|---|---|
| observedValue | A String. Indicates the value of the indicator which crossed the threshold. |
| threshold | A ThresholdRef. |
| thresholdCrossingDescription | A String. Indicates further information on the threshold crossing alarm. |

### RelatedPlace sub-resource fields

| | |
|---|---|
| relatedPlace | A Place. Place reference. |
| role | A String. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ServiceRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ThresholdRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Place sub-resource fields

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| | |
|---|---|
| externalIdentifier | An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ExternalIdentifier sub-resource fields

| | |
|---|---|
| externalIdentifierType | A String. Type of the identification, typically would be the type of the entity within the external system. |
| id | A String. Identification of the entity within the external system. |
| owner | A String. Name of the external system that owns the entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Json representation sample(s)

We provide below a JSON representation as example of the 'UnAckAlarm' resource object.


```
{
    "id": "1052",
    "@type": "UnAckAlarm",
    "href": "/alarmManagement/v5/unAckAlarm/1052",
    "ackSystemId": "ems-1",
    "ackUserId": "bob@example.net",
    "ackTime": "2019-07-03T03:38:06.890Z",
    "alarmPattern": [
        {
            "@type": "Alarm",
            "id": "8675309"
        },
        {
            "@type": "Alarm",
```


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```
            "id": "8675399"
        }
    ],
    "state": "done",
    "unAckedAlarm": [
        {
            "id": "8675309",
            "href": "/alarmManagement/v5/alarm/8675309",
            "@type": "Alarm",
            "externalAlarmId": "5551212",
            "state": "raised",
            "alarmType": "environmentalAlarm",
            "perceivedSeverity": "minor",
            "probableCause": "rectifierLowVoltage",
            "specificProblem": "ps=3,sl=1,in=8",
            "alarmedObjectType": "Rectifier",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "93051825",
                "href": "/resourceInventoryManagement/v5/resource/93051825"
            },
            "sourceSystemId": "ems-1",
            "alarmDetails": "voltage=204",
            "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
            "alarmReportingTime": "2019-07-03T03:32:17.552Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackState": "unacknowledged",
            "correlatedAlarm": [
                {
                    "@type": "AlarmRef",
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/868675399"
                }
            ],
            "comment": [
                {
                    "@type": "Comment",
                    "userId": "bob@example.net",
                    "systemId": "ems-1",
                    "time": "2019-07-03T03:37:33.827Z",
                    "comment": "Dispatched"
                }
            ]
        },
        {
            "id": "8675399",
            "href": "/alarmManagement/v5/alarm/8675399",
            "@type": "Alarm",
            "@schemaLocation": "/alarmManagement/v5//Alarm.schema.json",
            "externalAlarmId": "5551234",
            "state": "raised",
            "alarmType": "equipmentAlarm",
            "perceivedSeverity": "minor",
            "probableCause": "powerSupplyFailure",
            "specificProblem": "ch=1,ps=3",
            "alarmedObjectType": "Chassis",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "95830731",
                "href": "/resourceInventoryManagement/v5/resource/95830731"
```


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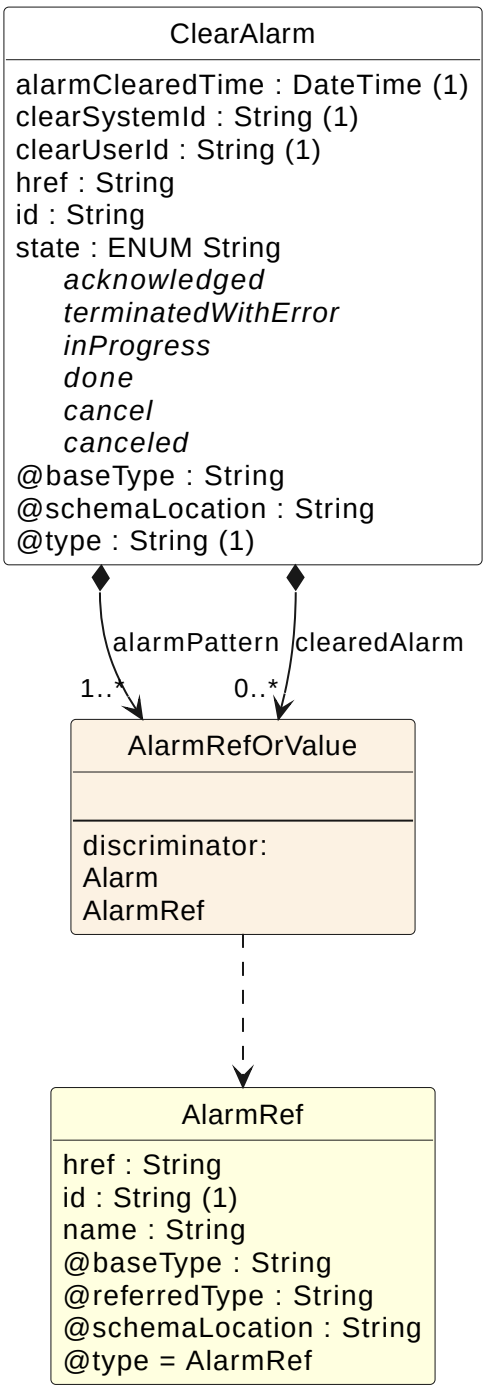
```
            },
            "sourceSystemId": "ems-1",
            "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
            "alarmReportingTime": "2019-07-03T03:32:18.102Z",
            "alarmChangedTime": "2019-07-03T03:32:52.744Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackState": "unacknowledged",
            "parentAlarm": [
                {
                    "@type": "AlarmRef",
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/868675309"
                }
            ],
            "comment": [
                {
                    "@type": "Comment",
                    "userId": "bob@example.net",
                    "systemId": "ems-1",
                    "time": "2019-07-03T03:37:33.827Z",
                    "comment": "Dispatched"
                }
            ]
        }
    ]
}
```


## ClearAlarm resource

Task resource for clear alarms operation.

### Resource model

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| Color | Description |
|---|---|
| | Resource (entry point) |
| | Sub-resource |
| | Sub-resource (reference entity) |

(1) : Mandatory property

*Figure 7 - ClearAlarm*

[.] | .//

*Figure 8 - ClearAlarm*

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## Field descriptions

### ClearAlarm fields

| | |
|---|---|
| alarmClearedTime | A DateTime. Time of the alarm clearing. |
| alarmPattern | An AlarmRefOrValue. An alarm defined by reference or value. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Alarm entity and not the AlarmRefOrValue class itself. |
| clearSystemId | A String. Name of the clearing system. |
| clearUserId | A String. Name of the clearing user. |
| clearedAlarm | An AlarmRefOrValue. An alarm defined by reference or value. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Alarm entity and not the AlarmRefOrValue class itself. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| state | A String. Current state of the operation task. ENUMERATED with values:<br>* acknowledged<br>* terminatedWithError<br>* inProgress<br>* done<br>* cancel<br>* canceled |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Alarm sub-resource fields

| | |
|---|---|
| ackState | A String. Provides the Acknowledgement State of the alarm (unacknowledged, acknowledged). ENUMERATED with values:<br>* unacknowledged<br>* acknowledged |
| ackSystemId | A String. Provides the name of the system that last changed the ackState of an alarm, i.e. acknowledged or unacknowledged the alarm. |
| ackUserId | A String. Provides the id of the user who has last changed the ack state of the alarm, i.e. acknowledged or unacknowledged the alarm. |
| affectedService | A ServiceRef. Service reference, for when Service is used by other entities. |

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| | |
|---|---|
| alarmChangedTime | A DateTime. Indicates the last date and time when the alarm is changed on the alarm-owning system. Any change to the alarm whether coming from the alarmed resource, or triggered by a change from the client is changing this time. |
| alarmClearedTime | A DateTime. Indicates the time (as a date + time) at which the alarm is cleared at the source. |
| alarmDetails | A String. Contains further information on the alarm. |
| alarmEscalation | A Boolean. Indicates if this alarm has been escalated or not. |
| alarmRaisedTime | A DateTime. Indicates the time (as a date + time) at which the alarm occurred at its source. |
| alarmReportingTime | A DateTime. Indicates the time (as a date + time) at which the alarm was reported by the owning OSS. It might be different from the alarmRaisedTime. For instance, if the alarm list is maintained by an EMS, the alarmRaisedtime would be the time the alarm was detected by the NE, while the alarmReportingTime would be the time this alarm was stored in the alarm list of the EMS. |
| alarmType | An AlarmType. Categorizes the alarm (X.733 8.1.1, 3GPP TS 32.111-2 Annex A). ENUMERATED with values:<br>* communicationsAlarm<br>* processingErrorAlarm<br>* environmentalAlarm<br>* qualityOfServiceAlarm<br>* equipmentAlarm<br>* integrityViolation<br>* operationalViolation<br>* physicalViolation<br>* securityService<br>* mechanismViolation<br>* timeDomainViolation |
| alarmedObject | An AlarmedObjectRef. Reference to object which affected by Alarm (AlarmedObject). |
| alarmedObjectType | A String. The type (class) of the managed object associated with the event. |
| clearSystemId | A String. Provides the id of the system where the user who invoked the alarmCleared operation is located. |
| clearUserId | A String. Provides the id of the user who invoked the alarmCleared operation. |
| comment | A Comment. Indicates the comments entered on the alarm. |
| correlatedAlarm | An AlarmRef. |
| crossedThresholdInformation | A CrossedThresholdInformation. Identifies the details of the threshold that has been crossed. |
| externalAlarmId | A String. An identifier of the alarm in the source system. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| isRootCause | A Boolean. Indicates whether the alarm is a root cause alarm.. |

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| | |
|---|---|
| parentAlarm | An AlarmRef. |
| perceivedSeverity | A PerceivedSeverity. Lists the possible severities that can be allocated to an Alarm. The values are consistent with ITU-T Recommendation X.733. Once an alarm has been cleared, its perceived severity is set to 'cleared' and can no longer be set. ENUMERATED with values:<br>* critical<br>* major<br>* minor<br>* warning<br>* indeterminate<br>* cleared |
| place | A RelatedPlace. A Place and an associated role as installation address, delivery address, etc.… |
| plannedOutageIndicator | A String. Indicates that the Managed Object (related to this alarm) is in planned outage (in planned maintenance, or out-of-service). ENUMERATED with values:<br>* InService<br>* OutOfService |
| probableCause | A String. Provides the probable cause of the alarm. The values are consistent with ITU-T Recommendation X.733 or 3GPP TS 32.111-2 Annex B. |
| proposedRepairedActions | A String. Indicates proposed repair actions, if known to the system emitting the alarm. |
| reportingSystemId | A String. Reporting system identity. |
| serviceAffecting | A Boolean. Indicates whether the alarm affects service or not. |
| sourceSystemId | A String. Source system identity. |
| specificProblem | A String. Provides more specific information about the alarm. |
| state | A String. Defines the alarm state during its life cycle (raised, updated, cleared). ENUMERATED with values:<br>* raised<br>* updated<br>* cleared |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### AlarmRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |

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| | |
|---|---|
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Comment sub-resource fields

| | |
|---|---|
| comment | A String. Indicates the text of the comment. |
| systemId | A String. Indicates the system identifier on which the client set the comment. |
| time | A DateTime. Indicates the time commenting the alarm. |
| userId | A String. Indicates the user commenting the alarm. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### AlarmedObjectRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### CrossedThresholdInformation sub-resource fields

| | |
|---|---|
| direction | A String. Indicates the threshold crossing direction: up or down. |
| granularity | A String. Indicates the granularity at which the indicator is evaluated for threshold crossing. |
| indicatorName | A String. Indicates the name of indicator which crossed the threshold. |
| indicatorUnit | A String. Indicates the unit of the measurement of the indicator corresponding to the threshold that has been crossed. |

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| | |
|---|---|
| observedValue | A String. Indicates the value of the indicator which crossed the threshold. |
| threshold | A ThresholdRef. |
| thresholdCrossingDescription | A String. Indicates further information on the threshold crossing alarm. |

### RelatedPlace sub-resource fields

| | |
|---|---|
| relatedPlace | A Place. Place reference. |
| role | A String. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ServiceRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ThresholdRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Place sub-resource fields

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| | |
|---|---|
| externalIdentifier | An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ExternalIdentifier sub-resource fields

| | |
|---|---|
| externalIdentifierType | A String. Type of the identification, typically would be the type of the entity within the external system. |
| id | A String. Identification of the entity within the external system. |
| owner | A String. Name of the external system that owns the entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Json representation sample(s)

We provide below a JSON representation as example of the 'ClearAlarm' resource object.


```
{
    "id": "1089",
    "href": "/alarmManagement/v5/clearAlarm/1089",
    "@type": "ClearAlarm",
    "clearSystemId": "ems-1",
    "clearUserId": "bob@example.net",
    "alarmClearedTime": "2019-07-03T03:55:45.937Z",
    "alarmPattern": [
        {
            "@type": "Alarm",
            "id": "8675309"
        },
        {
            "@type": "Alarm",
```


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```
            "id": "8675399"
        }
    ],
    "state": "done",
    "clearedAlarm": [
        {
            "id": "8675309",
            "href": "/alarmManagement/v5/alarm/8675309",
            "@type": "Alarm",
            "externalAlarmId": "5551212",
            "state": "cleared",
            "alarmType": "environmentalAlarm",
            "perceivedSeverity": "cleared",
            "probableCause": "rectifierLowVoltage",
            "specificProblem": "ps=3,sl=1,in=8",
            "alarmedObjectType": "Rectifier",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "93051825",
                "href": "/resourceInventoryManagement/v5/resource/93051825"
            },
            "sourceSystemId": "ems-1",
            "alarmDetails": "voltage=204",
            "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
            "alarmReportingTime": "2019-07-03T03:32:17.552Z",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackState": "unacknowledged",
            "comment": [
                {
                    "@type": "Comment",
                    "userId": "bob@example.net",
                    "systemId": "ems-1",
                    "time": "2019-07-03T03:37:33.827Z",
                    "comment": "Dispatched"
                }
            ],
            "clearSystemId": "ems-1",
            "clearUserId": "bob@example.net",
            "alarmClearedTime": "2019-07-03T03:55:45.937Z"
        },
        {
            "id": "8675399",
            "href": "/alarmManagement/v5/alarm/8675399",
            "@type": "Alarm",
            "externalAlarmId": "5551234",
            "state": "cleared",
            "alarmType": "equipmentAlarm",
            "perceivedSeverity": "cleared",
            "probableCause": "powerSupplyFailure",
            "specificProblem": "ch=1,ps=3",
            "alarmedObjectType": "Chassis",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "95830731",
                "href": "/resourceInventoryManagement/v5/resource/95830731"
            },
            "sourceSystemId": "ems-1",
            "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
            "alarmReportingTime": "2019-07-03T03:32:18.102Z",
```


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```
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackState": "unacknowledged",
            "comment": [
                {
                    "@type": "Comment",
                    "userId": "bob@example.net",
                    "systemId": "ems-1",
                    "time": "2019-07-03T03:37:33.827Z",
                    "comment": "Dispatched"
                }
            ],
            "clearSystemId": "ems-1",
            "clearUserId": "bob@example.net",
            "alarmClearedTime": "2019-07-03T03:55:45.937Z"
        }
    ]
}
```


## CommentAlarm resource

Task resource for comment alarms operation.

### Resource model

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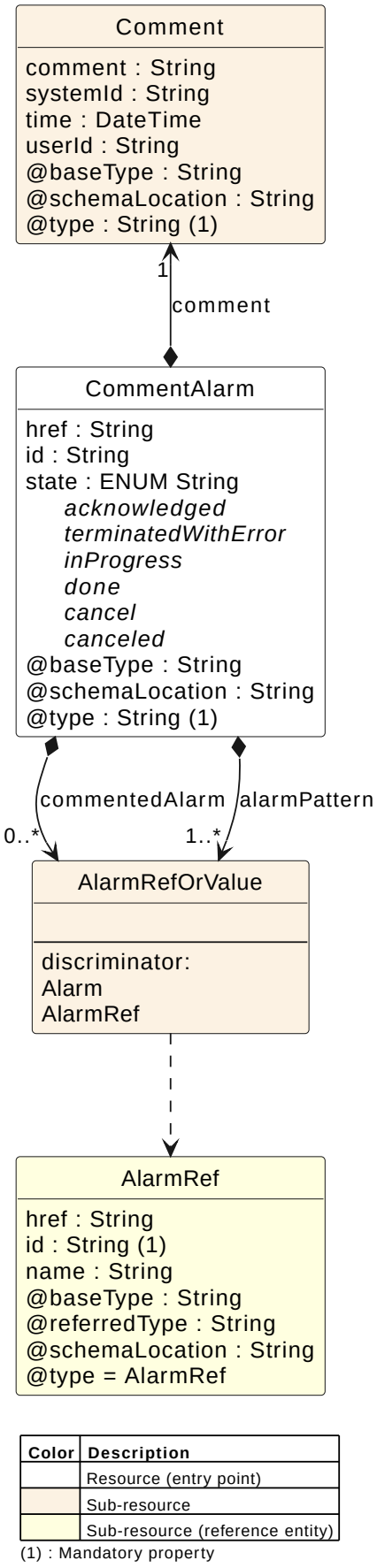



*Figure 9 - CommentAlarm*

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[.] | .//

*Figure 10 - CommentAlarm*

### Field descriptions

### CommentAlarm fields

| | |
|---|---|
| alarmPattern | An AlarmRefOrValue. An alarm defined by reference or value. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Alarm entity and not the AlarmRefOrValue class itself. |
| comment | A Comment. Indicates the comments entered on the alarm. |
| commentedAlarm | An AlarmRefOrValue. An alarm defined by reference or value. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Alarm entity and not the AlarmRefOrValue class itself. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| state | A String. Current state of the operation task. ENUMERATED with values:<br>* acknowledged<br>* terminatedWithError<br>* inProgress<br>* done<br>* cancel<br>* canceled |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Comment sub-resource fields

| | |
|---|---|
| comment | A String. Indicates the text of the comment. |
| systemId | A String. Indicates the system identifier on which the client set the comment. |
| time | A DateTime. Indicates the time commenting the alarm. |
| userId | A String. Indicates the user commenting the alarm. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Alarm sub-resource fields

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| | |
|---|---|
| ackState | A String. Provides the Acknowledgement State of the alarm (unacknowledged, acknowledged). ENUMERATED with values:<br>* unacknowledged<br>* acknowledged |
| ackSystemId | A String. Provides the name of the system that last changed the ackState of an alarm, i.e. acknowledged or unacknowledged the alarm. |
| ackUserId | A String. Provides the id of the user who has last changed the ack state of the alarm, i.e. acknowledged or unacknowledged the alarm. |
| affectedService | A ServiceRef. Service reference, for when Service is used by other entities. |
| alarmChangedTime | A DateTime. Indicates the last date and time when the alarm is changed on the alarm-owning system. Any change to the alarm whether coming from the alarmed resource, or triggered by a change from the client is changing this time. |
| alarmClearedTime | A DateTime. Indicates the time (as a date + time) at which the alarm is cleared at the source. |
| alarmDetails | A String. Contains further information on the alarm. |
| alarmEscalation | A Boolean. Indicates if this alarm has been escalated or not. |
| alarmRaisedTime | A DateTime. Indicates the time (as a date + time) at which the alarm occurred at its source. |
| alarmReportingTime | A DateTime. Indicates the time (as a date + time) at which the alarm was reported by the owning OSS. It might be different from the alarmRaisedTime. For instance, if the alarm list is maintained by an EMS, the alarmRaisedtime would be the time the alarm was detected by the NE, while the alarmReportingTime would be the time this alarm was stored in the alarm list of the EMS. |
| alarmType | An AlarmType. Categorizes the alarm (X.733 8.1.1, 3GPP TS 32.111-2 Annex A). ENUMERATED with values:<br>* communicationsAlarm<br>* processingErrorAlarm<br>* environmentalAlarm<br>* qualityOfServiceAlarm<br>* equipmentAlarm<br>* integrityViolation<br>* operationalViolation<br>* physicalViolation<br>* securityService<br>* mechanismViolation<br>* timeDomainViolation |
| alarmedObject | An AlarmedObjectRef. Reference to object which affected by Alarm (AlarmedObject). |
| alarmedObjectType | A String. The type (class) of the managed object associated with the event. |
| clearSystemId | A String. Provides the id of the system where the user who invoked the alarmCleared operation is located. |

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| | |
|---|---|
| clearUserId | A String. Provides the id of the user who invoked the alarmCleared operation. |
| comment | A Comment. Indicates the comments entered on the alarm. |
| correlatedAlarm | An AlarmRef. |
| crossedThresholdInformation | A CrossedThresholdInformation. Identifies the details of the threshold that has been crossed. |
| externalAlarmId | A String. An identifier of the alarm in the source system. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| isRootCause | A Boolean. Indicates whether the alarm is a root cause alarm.. |
| parentAlarm | An AlarmRef. |
| perceivedSeverity | A PerceivedSeverity. Lists the possible severities that can be allocated to an Alarm. The values are consistent with ITU-T Recommendation X.733. Once an alarm has been cleared, its perceived severity is set to 'cleared' and can no longer be set. ENUMERATED with values:<br>* critical<br>* major<br>* minor<br>* warning<br>* indeterminate<br>* cleared |
| place | A RelatedPlace. A Place and an associated role as installation address, delivery address, etc.… |
| plannedOutageIndicator | A String. Indicates that the Managed Object (related to this alarm) is in planned outage (in planned maintenance, or out-of-service). ENUMERATED with values:<br>* InService<br>* OutOfService |
| probableCause | A String. Provides the probable cause of the alarm. The values are consistent with ITU-T Recommendation X.733 or 3GPP TS 32.111-2 Annex B. |
| proposedRepairedActions | A String. Indicates proposed repair actions, if known to the system emitting the alarm. |
| reportingSystemId | A String. Reporting system identity. |
| serviceAffecting | A Boolean. Indicates whether the alarm affects service or not. |
| sourceSystemId | A String. Source system identity. |
| specificProblem | A String. Provides more specific information about the alarm. |
| state | A String. Defines the alarm state during its life cycle (raised, updated, cleared). ENUMERATED with values:<br>* raised<br>* updated<br>* cleared |
| @baseType | A String. When sub-classing, this defines the super-class. |

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| | |
|---|---|
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### AlarmRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### AlarmedObjectRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### CrossedThresholdInformation sub-resource fields

| | |
|---|---|
| direction | A String. Indicates the threshold crossing direction: up or down. |
| granularity | A String. Indicates the granularity at which the indicator is evaluated for threshold crossing. |
| indicatorName | A String. Indicates the name of indicator which crossed the threshold. |
| indicatorUnit | A String. Indicates the unit of the measurement of the indicator corresponding to the threshold that has been crossed. |
| observedValue | A String. Indicates the value of the indicator which crossed the threshold. |
| threshold | A ThresholdRef. |

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| | |
|---|---|
| thresholdCrossingDescription | A String. Indicates further information on the threshold crossing alarm. |

### RelatedPlace sub-resource fields

| | |
|---|---|
| relatedPlace | A Place. Place reference. |
| role | A String. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ServiceRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ThresholdRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Place sub-resource fields

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| | |
|---|---|
| externalIdentifier | An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ExternalIdentifier sub-resource fields

| | |
|---|---|
| externalIdentifierType | A String. Type of the identification, typically would be the type of the entity within the external system. |
| id | A String. Identification of the entity within the external system. |
| owner | A String. Name of the external system that owns the entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Json representation sample(s)

We provide below a JSON representation as example of the 'CommentAlarm' resource object.


```
{
    "id": "1045",
    "@type": "CommentAlarm",
    "href": "/alarmManagement/v5/commentAlarm/1045",
    "comment": {
        "@type": "Comment",
        "userId": "bob@example.net",
        "systemId": "ems-1",
        "time": "2019-07-03T03:37:33.827Z",
        "comment": "Dispatched"
    },
    "alarmPattern": [
        {
```


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```
            "@type": "Alarm",
            "id": "8675309"
        },
        {
            "@type": "Alarm",
            "id": "8675399"
        }
    ],
    "state": "done",
    "commentedAlarm": [
        {
            "id": "8675309",
            "href": "/alarmManagement/v5/alarm/8675309",
            "@type": "Alarm",
            "externalAlarmId": "5551212",
            "state": "raised",
            "alarmType": "environmentalAlarm",
            "perceivedSeverity": "minor",
            "probableCause": "rectifierLowVoltage",
            "specificProblem": "ps=3,sl=1,in=8",
            "alarmedObjectType": "Rectifier",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "93051825",
                "href": "/resourceInventoryManagement/v5/resource/93051825"
            },
            "sourceSystemId": "ems-1",
            "alarmDetails": "voltage=204",
            "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
            "alarmReportingTime": "2019-07-03T03:32:17.552Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackState": "acknowledged",
            "correlatedAlarm": [
                {
                    "@type": "AlarmRef",
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/868675399"
                }
            ],
            "comment": [
                {
                    "@type": "Comment",
                    "userId": "bob@example.net",
                    "systemId": "ems-1",
                    "time": "2019-07-03T03:37:33.827Z",
                    "comment": "Dispatched"
                }
            ]
        },
        {
            "id": "8675399",
            "href": "/alarmManagement/v5/alarm/8675399",
            "@type": "Alarm",
            "externalAlarmId": "5551234",
            "state": "raised",
            "alarmType": "equipmentAlarm",
            "perceivedSeverity": "minor",
            "probableCause": "powerSupplyFailure",
            "specificProblem": "ch=1,ps=3",
            "alarmedObjectType": "Chassis",
            "alarmedObject": {
```


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```
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "95830731",
                "href": "/resourceInventoryManagement/v5/resource/95830731"
            },
            "sourceSystemId": "ems-1",
            "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
            "alarmReportingTime": "2019-07-03T03:32:18.102Z",
            "alarmChangedTime": "2019-07-03T03:32:52.744Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackState": "acknowledged",
            "parentAlarm": [
                {
                    "@type": "AlarmRef",
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/868675309"
                }
            ],
            "comment": [
                {
                    "@type": "Comment",
                    "userId": "bob@example.net",
                    "systemId": "ems-1",
                    "time": "2019-07-03T03:37:33.827Z",
                    "comment": "Dispatched"
                }
            ]
        }
    ]
}
```


## GroupAlarm resource

Task resource for group alarms operation.

### Resource model

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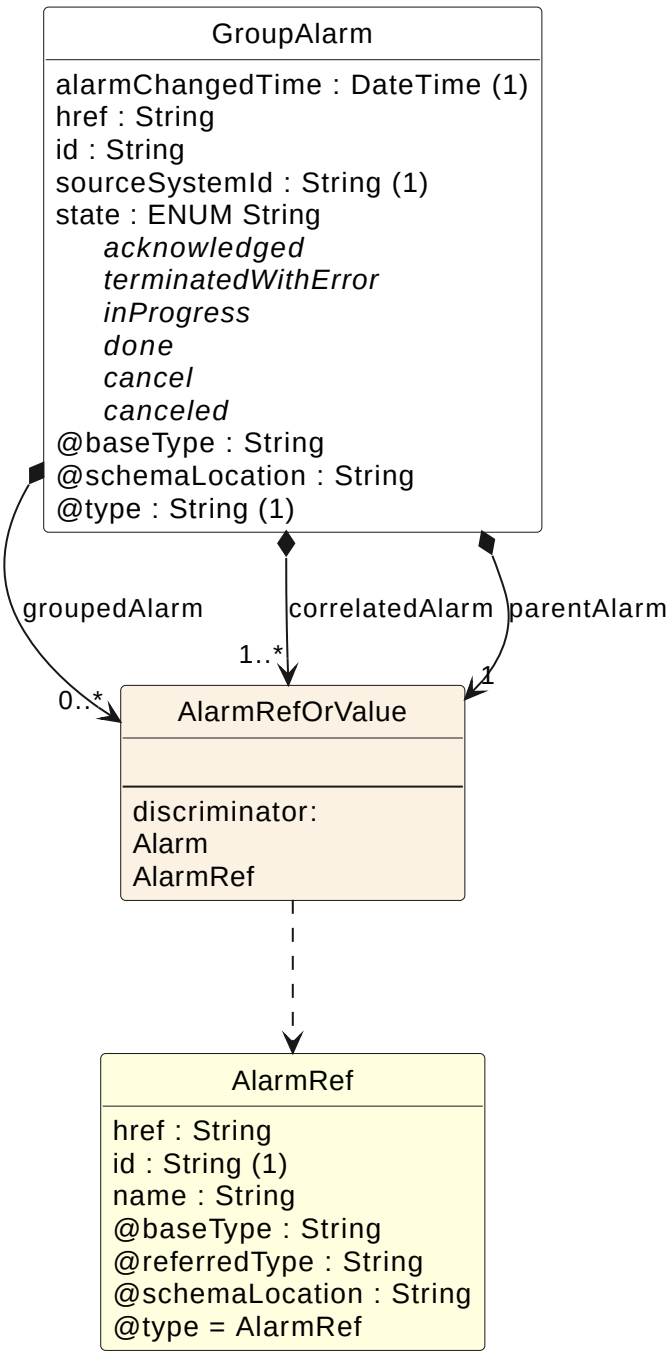



*Figure 11 - GroupAlarm*

[.] | .//

*Figure 12 - GroupAlarm*

## Field descriptions

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### GroupAlarm fields

| | |
|---|---|
| alarmChangedTime | A DateTime. Time of the correlation. |
| correlatedAlarm | An AlarmRefOrValue. An alarm defined by reference or value. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Alarm entity and not the AlarmRefOrValue class itself. |
| groupedAlarm | An AlarmRefOrValue. An alarm defined by reference or value. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Alarm entity and not the AlarmRefOrValue class itself. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| parentAlarm | An AlarmRefOrValue. An alarm defined by reference or value. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Alarm entity and not the AlarmRefOrValue class itself. |
| sourceSystemId | A String. Source system identifier. |
| state | A String. Current state of the operation task. ENUMERATED with values:<br>* acknowledged<br>* terminatedWithError<br>* inProgress<br>* done<br>* cancel<br>* canceled |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Alarm sub-resource fields

| | |
|---|---|
| ackState | A String. Provides the Acknowledgement State of the alarm (unacknowledged, acknowledged). ENUMERATED with values:<br>* unacknowledged<br>* acknowledged |
| ackSystemId | A String. Provides the name of the system that last changed the ackState of an alarm, i.e. acknowledged or unacknowledged the alarm. |
| ackUserId | A String. Provides the id of the user who has last changed the ack state of the alarm, i.e. acknowledged or unacknowledged the alarm. |
| affectedService | A ServiceRef. Service reference, for when Service is used by other entities. |

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| | |
|---|---|
| alarmChangedTime | A DateTime. Indicates the last date and time when the alarm is changed on the alarm-owning system. Any change to the alarm whether coming from the alarmed resource, or triggered by a change from the client is changing this time. |
| alarmClearedTime | A DateTime. Indicates the time (as a date + time) at which the alarm is cleared at the source. |
| alarmDetails | A String. Contains further information on the alarm. |
| alarmEscalation | A Boolean. Indicates if this alarm has been escalated or not. |
| alarmRaisedTime | A DateTime. Indicates the time (as a date + time) at which the alarm occurred at its source. |
| alarmReportingTime | A DateTime. Indicates the time (as a date + time) at which the alarm was reported by the owning OSS. It might be different from the alarmRaisedTime. For instance, if the alarm list is maintained by an EMS, the alarmRaisedtime would be the time the alarm was detected by the NE, while the alarmReportingTime would be the time this alarm was stored in the alarm list of the EMS. |
| alarmType | An AlarmType. Categorizes the alarm (X.733 8.1.1, 3GPP TS 32.111-2 Annex A). ENUMERATED with values:<br>* communicationsAlarm<br>* processingErrorAlarm<br>* environmentalAlarm<br>* qualityOfServiceAlarm<br>* equipmentAlarm<br>* integrityViolation<br>* operationalViolation<br>* physicalViolation<br>* securityService<br>* mechanismViolation<br>* timeDomainViolation |
| alarmedObject | An AlarmedObjectRef. Reference to object which affected by Alarm (AlarmedObject). |
| alarmedObjectType | A String. The type (class) of the managed object associated with the event. |
| clearSystemId | A String. Provides the id of the system where the user who invoked the alarmCleared operation is located. |
| clearUserId | A String. Provides the id of the user who invoked the alarmCleared operation. |
| comment | A Comment. Indicates the comments entered on the alarm. |
| correlatedAlarm | An AlarmRef. |
| crossedThresholdInformation | A CrossedThresholdInformation. Identifies the details of the threshold that has been crossed. |
| externalAlarmId | A String. An identifier of the alarm in the source system. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| isRootCause | A Boolean. Indicates whether the alarm is a root cause alarm.. |

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| | |
|---|---|
| parentAlarm | An AlarmRef. |
| perceivedSeverity | A PerceivedSeverity. Lists the possible severities that can be allocated to an Alarm. The values are consistent with ITU-T Recommendation X.733. Once an alarm has been cleared, its perceived severity is set to 'cleared' and can no longer be set. ENUMERATED with values:<br>* critical<br>* major<br>* minor<br>* warning<br>* indeterminate<br>* cleared |
| place | A RelatedPlace. A Place and an associated role as installation address, delivery address, etc.… |
| plannedOutageIndicator | A String. Indicates that the Managed Object (related to this alarm) is in planned outage (in planned maintenance, or out-of-service). ENUMERATED with values:<br>* InService<br>* OutOfService |
| probableCause | A String. Provides the probable cause of the alarm. The values are consistent with ITU-T Recommendation X.733 or 3GPP TS 32.111-2 Annex B. |
| proposedRepairedActions | A String. Indicates proposed repair actions, if known to the system emitting the alarm. |
| reportingSystemId | A String. Reporting system identity. |
| serviceAffecting | A Boolean. Indicates whether the alarm affects service or not. |
| sourceSystemId | A String. Source system identity. |
| specificProblem | A String. Provides more specific information about the alarm. |
| state | A String. Defines the alarm state during its life cycle (raised, updated, cleared). ENUMERATED with values:<br>* raised<br>* updated<br>* cleared |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### AlarmRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |

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| | |
|---|---|
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Comment sub-resource fields

| | |
|---|---|
| comment | A String. Indicates the text of the comment. |
| systemId | A String. Indicates the system identifier on which the client set the comment. |
| time | A DateTime. Indicates the time commenting the alarm. |
| userId | A String. Indicates the user commenting the alarm. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### AlarmedObjectRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### CrossedThresholdInformation sub-resource fields

| | |
|---|---|
| direction | A String. Indicates the threshold crossing direction: up or down. |
| granularity | A String. Indicates the granularity at which the indicator is evaluated for threshold crossing. |
| indicatorName | A String. Indicates the name of indicator which crossed the threshold. |
| indicatorUnit | A String. Indicates the unit of the measurement of the indicator corresponding to the threshold that has been crossed. |

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| | |
|---|---|
| observedValue | A String. Indicates the value of the indicator which crossed the threshold. |
| threshold | A ThresholdRef. |
| thresholdCrossingDescription | A String. Indicates further information on the threshold crossing alarm. |

### RelatedPlace sub-resource fields

| | |
|---|---|
| relatedPlace | A Place. Place reference. |
| role | A String. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ServiceRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ThresholdRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Place sub-resource fields

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| | |
|---|---|
| externalIdentifier | An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ExternalIdentifier sub-resource fields

| | |
|---|---|
| externalIdentifierType | A String. Type of the identification, typically would be the type of the entity within the external system. |
| id | A String. Identification of the entity within the external system. |
| owner | A String. Name of the external system that owns the entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Json representation sample(s)

We provide below a JSON representation as example of the 'GroupAlarm' resource object.


```
{
    "id": "1029",
    "@type": "GroupAlarm",
    "href": "/alarmManagement/v5/groupAlarm/1029",
    "sourceSystemId": "ems-1",
    "alarmChangedTime": "2019-07-03T03:32:52.744Z",
    "parentAlarm": {
        "@type": "Alarm",
        "id": "8675309"
    },
    "correlatedAlarm": [
        {
            "@type": "Alarm",
            "id": "8675399"
```


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```
        }
    ],
    "state": "done",
    "groupedAlarm": [
        {
            "id": "8675309",
            "href": "/alarmManagement/v5/alarm/8675309",
            "@type": "Alarm",
            "externalAlarmId": "5551212",
            "state": "raised",
            "alarmType": "environmentalAlarm",
            "perceivedSeverity": "minor",
            "probableCause": "rectifierLowVoltage",
            "specificProblem": "ps=3,sl=1,in=8",
            "alarmedObjectType": "Rectifier",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "93051825",
                "href": "/resourceInventoryManagement/v5/resource/93051825"
            },
            "sourceSystemId": "ems-1",
            "alarmDetails": "voltage=204",
            "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
            "alarmReportingTime": "2019-07-03T03:32:17.552Z",
            "alarmChangedTime": "2019-07-03T03:32:52.744Z",
            "correlatedAlarm": [
                {
                    "@type": "AlarmRef",
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/868675399"
                }
            ]
        },
        {
            "id": "8675399",
            "href": "/alarmManagement/v5/alarm/8675399",
            "@type": "Alarm",
            "@schemaLocation": "/alarmManagement/v5/Alarm.schema.json",
            "externalAlarmId": "5551234",
            "state": "raised",
            "alarmType": "equipmentAlarm",
            "perceivedSeverity": "minor",
            "probableCause": "powerSupplyFailure",
            "specificProblem": "ch=1,ps=3",
            "alarmedObjectType": "Chassis",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "95830731",
                "href": "/resourceInventoryManagement/v5/resource/95830731"
            },
            "sourceSystemId": "ems-1",
            "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
            "alarmReportingTime": "2019-07-03T03:32:18.102Z",
            "alarmChangedTime": "2019-07-03T03:32:52.744Z",
            "parentAlarm": [
                {
                    "@type": "AlarmRef",
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/868675309"
                }
            ]
```


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```
        }
    ]
}
```

## UnGroupAlarm resource

Task resource for ungroup alarms operation.

### Resource model

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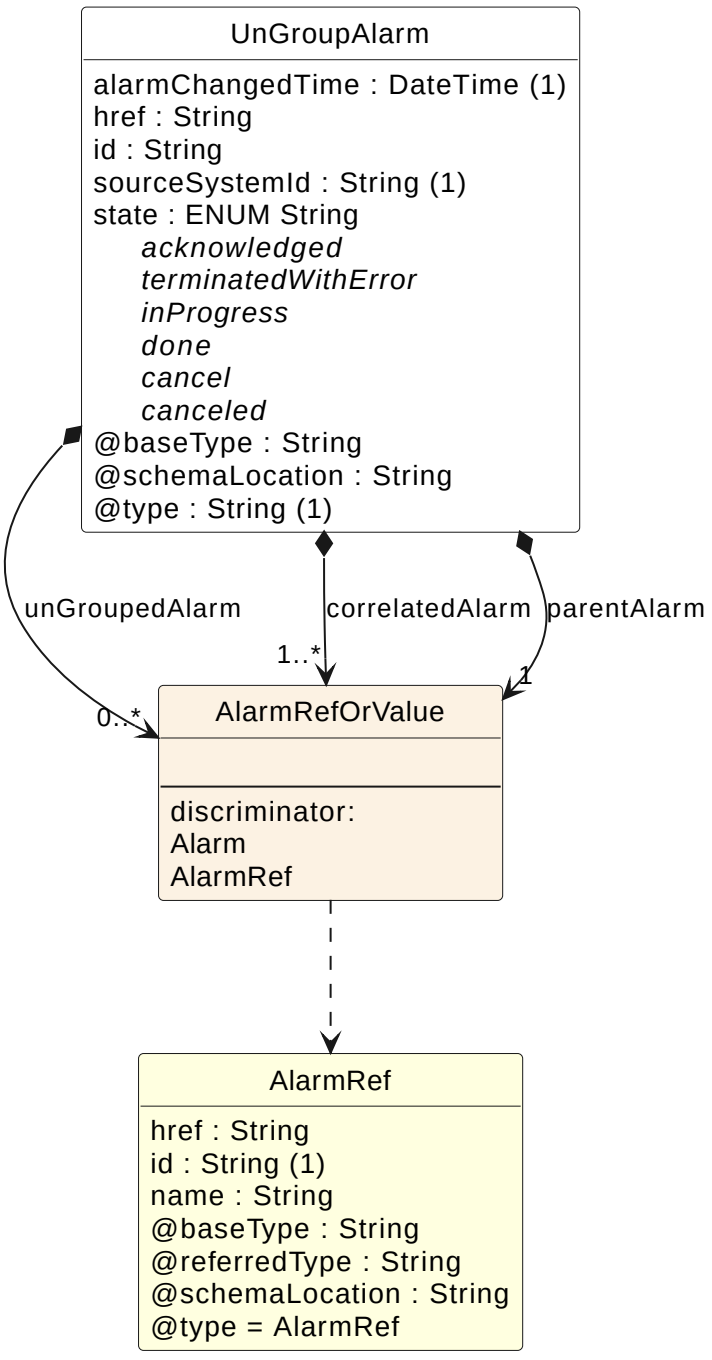



*Figure 13 - UnGroupAlarm*

[.] | .//

*Figure 14 - UnGroupAlarm*

## Field descriptions

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### UnGroupAlarm fields

| | |
|---|---|
| alarmChangedTime | A DateTime. Time of the uncorrelation. |
| correlatedAlarm | An AlarmRefOrValue. An alarm defined by reference or value. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Alarm entity and not the AlarmRefOrValue class itself. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| parentAlarm | An AlarmRefOrValue. An alarm defined by reference or value. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Alarm entity and not the AlarmRefOrValue class itself. |
| sourceSystemId | A String. Source system identifier. |
| state | A String. Current state of the operation task. ENUMERATED with values:<br>* acknowledged<br>* terminatedWithError<br>* inProgress<br>* done<br>* cancel<br>* canceled |
| unGroupedAlarm | An AlarmRefOrValue. An alarm defined by reference or value. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Alarm entity and not the AlarmRefOrValue class itself. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Alarm sub-resource fields

| | |
|---|---|
| ackState | A String. Provides the Acknowledgement State of the alarm (unacknowledged, acknowledged). ENUMERATED with values:<br>* unacknowledged<br>* acknowledged |
| ackSystemId | A String. Provides the name of the system that last changed the ackState of an alarm, i.e. acknowledged or unacknowledged the alarm. |
| ackUserId | A String. Provides the id of the user who has last changed the ack state of the alarm, i.e. acknowledged or unacknowledged the alarm. |
| affectedService | A ServiceRef. Service reference, for when Service is used by other entities. |

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| | |
|---|---|
| alarmChangedTime | A DateTime. Indicates the last date and time when the alarm is changed on the alarm-owning system. Any change to the alarm whether coming from the alarmed resource, or triggered by a change from the client is changing this time. |
| alarmClearedTime | A DateTime. Indicates the time (as a date + time) at which the alarm is cleared at the source. |
| alarmDetails | A String. Contains further information on the alarm. |
| alarmEscalation | A Boolean. Indicates if this alarm has been escalated or not. |
| alarmRaisedTime | A DateTime. Indicates the time (as a date + time) at which the alarm occurred at its source. |
| alarmReportingTime | A DateTime. Indicates the time (as a date + time) at which the alarm was reported by the owning OSS. It might be different from the alarmRaisedTime. For instance, if the alarm list is maintained by an EMS, the alarmRaisedtime would be the time the alarm was detected by the NE, while the alarmReportingTime would be the time this alarm was stored in the alarm list of the EMS. |
| alarmType | An AlarmType. Categorizes the alarm (X.733 8.1.1, 3GPP TS 32.111-2 Annex A). ENUMERATED with values:<br>* communicationsAlarm<br>* processingErrorAlarm<br>* environmentalAlarm<br>* qualityOfServiceAlarm<br>* equipmentAlarm<br>* integrityViolation<br>* operationalViolation<br>* physicalViolation<br>* securityService<br>* mechanismViolation<br>* timeDomainViolation |
| alarmedObject | An AlarmedObjectRef. Reference to object which affected by Alarm (AlarmedObject). |
| alarmedObjectType | A String. The type (class) of the managed object associated with the event. |
| clearSystemId | A String. Provides the id of the system where the user who invoked the alarmCleared operation is located. |
| clearUserId | A String. Provides the id of the user who invoked the alarmCleared operation. |
| comment | A Comment. Indicates the comments entered on the alarm. |
| correlatedAlarm | An AlarmRef. |
| crossedThresholdInformation | A CrossedThresholdInformation. Identifies the details of the threshold that has been crossed. |
| externalAlarmId | A String. An identifier of the alarm in the source system. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| isRootCause | A Boolean. Indicates whether the alarm is a root cause alarm.. |

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| | |
|---|---|
| parentAlarm | An AlarmRef. |
| perceivedSeverity | A PerceivedSeverity. Lists the possible severities that can be allocated to an Alarm. The values are consistent with ITU-T Recommendation X.733. Once an alarm has been cleared, its perceived severity is set to 'cleared' and can no longer be set. ENUMERATED with values:<br>* critical<br>* major<br>* minor<br>* warning<br>* indeterminate<br>* cleared |
| place | A RelatedPlace. A Place and an associated role as installation address, delivery address, etc.… |
| plannedOutageIndicator | A String. Indicates that the Managed Object (related to this alarm) is in planned outage (in planned maintenance, or out-of-service). ENUMERATED with values:<br>* InService<br>* OutOfService |
| probableCause | A String. Provides the probable cause of the alarm. The values are consistent with ITU-T Recommendation X.733 or 3GPP TS 32.111-2 Annex B. |
| proposedRepairedActions | A String. Indicates proposed repair actions, if known to the system emitting the alarm. |
| reportingSystemId | A String. Reporting system identity. |
| serviceAffecting | A Boolean. Indicates whether the alarm affects service or not. |
| sourceSystemId | A String. Source system identity. |
| specificProblem | A String. Provides more specific information about the alarm. |
| state | A String. Defines the alarm state during its life cycle (raised, updated, cleared). ENUMERATED with values:<br>* raised<br>* updated<br>* cleared |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### AlarmRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |

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| | |
|---|---|
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Comment sub-resource fields

| | |
|---|---|
| comment | A String. Indicates the text of the comment. |
| systemId | A String. Indicates the system identifier on which the client set the comment. |
| time | A DateTime. Indicates the time commenting the alarm. |
| userId | A String. Indicates the user commenting the alarm. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### AlarmedObjectRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### CrossedThresholdInformation sub-resource fields

| | |
|---|---|
| direction | A String. Indicates the threshold crossing direction: up or down. |
| granularity | A String. Indicates the granularity at which the indicator is evaluated for threshold crossing. |
| indicatorName | A String. Indicates the name of indicator which crossed the threshold. |
| indicatorUnit | A String. Indicates the unit of the measurement of the indicator corresponding to the threshold that has been crossed. |

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| | |
|---|---|
| observedValue | A String. Indicates the value of the indicator which crossed the threshold. |
| threshold | A ThresholdRef. |
| thresholdCrossingDescription | A String. Indicates further information on the threshold crossing alarm. |

### RelatedPlace sub-resource fields

| | |
|---|---|
| relatedPlace | A Place. Place reference. |
| role | A String. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ServiceRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ThresholdRef sub-resource fields

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Place sub-resource fields

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| | |
|---|---|
| externalIdentifier | An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### ExternalIdentifier sub-resource fields

| | |
|---|---|
| externalIdentifierType | A String. Type of the identification, typically would be the type of the entity within the external system. |
| id | A String. Identification of the entity within the external system. |
| owner | A String. Name of the external system that owns the entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String. When sub-classing, this defines the sub-class Extensible name. |

### Json representation sample(s)

We provide below a JSON representation as example of the 'UnGroupAlarm' resource object.


```
{
    "id": "1069",
    "href": "/alarmManagement/v5/unGroupAlarm/1069",
    "@type": "UnGroupAlarm",
    "sourceSystemId": "ems-1",
    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
    "parentAlarm": {
        "@type": "Alarm",
        "id": "8675309"
    },
    "correlatedAlarm": [
        {
            "@type": "Alarm",
```


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```
            "id": "8675399"
        },
        {
            "@type": "Alarm",
            "id": "8675399"
        }
    ],
    "state": "done",
    "unGroupedAlarm": [
        {
            "id": "8675309",
            "href": "/alarmManagement/v5/alarm/8675309",
            "@type": "Alarm",
            "externalAlarmId": "5551212",
            "state": "raised",
            "alarmType": "environmentalAlarm",
            "perceivedSeverity": "minor",
            "probableCause": "rectifierLowVoltage",
            "specificProblem": "ps=3,sl=1,in=8",
            "alarmedObjectType": "Rectifier",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "93051825",
                "href": "/resourceInventoryManagement/v5/resource/93051825"
            },
            "sourceSystemId": "ems-1",
            "alarmDetails": "voltage=204",
            "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
            "alarmReportingTime": "2019-07-03T03:32:17.552Z",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackState": "unacknowledged",
            "comment": [
                {
                    "@type": "Comment",
                    "userId": "bob@example.net",
                    "systemId": "ems-1",
                    "time": "2019-07-03T03:37:33.827Z",
                    "comment": "Dispatched"
                }
            ]
        },
        {
            "id": "8675399",
            "href": "/alarmManagement/v5/alarm/8675399",
            "@type": "Alarm",
            "externalAlarmId": "5551234",
            "state": "raised",
            "alarmType": "equipmentAlarm",
            "perceivedSeverity": "minor",
            "probableCause": "powerSupplyFailure",
            "specificProblem": "ch=1,ps=3",
            "alarmedObjectType": "Chassis",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "95830731",
                "href": "/resourceInventoryManagement/v5/resource/95830731"
            },
            "sourceSystemId": "ems-1",
            "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
```


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```
            "alarmReportingTime": "2019-07-03T03:32:18.102Z",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackState": "unacknowledged",
            "comment": [
                {
                    "@type": "Comment",
                    "userId": "bob@example.net",
                    "systemId": "ems-1",
                    "time": "2019-07-03T03:37:33.827Z",
                    "comment": "Dispatched"
                }
            ]
        }
    ]
}
```


## Notification Resource Models

16 notifications are defined for this API.

Notifications related to Alarm:

- Create Event
- Delete Event
- Attribute Value Change Event
- State Change Event

Notifications related to AckAlarm:

- Create Event
- State Change Event

Notifications related to UnAckAlarm:

- Create Event
- State Change Event

Notifications related to ClearAlarm:

- Create Event
- State Change Event

Notifications related to CommentAlarm:

- Create Event
- State Change Event

Notifications related to GroupAlarm:

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- Create Event
- State Change Event

Notifications related to UnGroupAlarm:

- Create Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

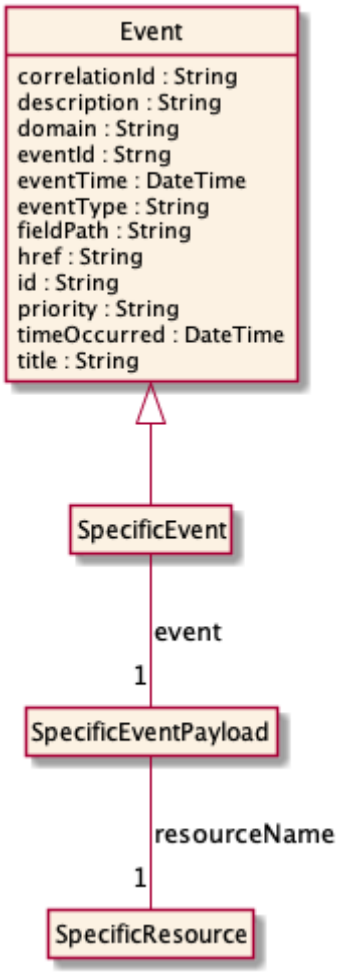



*Figure 15 Notification Pattern*

## Alarm

### Create Event

Message example for AlarmCreateEvent event

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```
Content-Type: application/json

{
    "correlationId": "96b37351-efde",
    "description": "UnGroupAlarmCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4505-85a4-b1a0c5587ebc",
    "eventTime": "2022-08-25T12:20:19.700Z",
    "eventType": "UnGroupAlarmCreateEvent",
    "priority": "3",
    "timeOccurred": "2022-08-25T12:20:10.499Z",
    "title": "UnGroupCreateEvent",
    "event": {
        "unGroupAlarm": {
            "@type": "UnGroupAlarm",
            "id": "1069",
            "href": "http://servername/unGroupAlarm/1069",
            "sourceSystemId": "ems-1",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "parentAlarm": {
                "@type": "Alarm",
                "id": "8675309"
            },
            "correlatedAlarm": [
                {
                    "@type": "Alarm",
                    "id": "8675399"
                },
                {
                    "@type": "Alarm",
                    "id": "8675399"
                }
            ],
            "state": "done",
            "unGroupedAlarm": [
                {
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/8675309",
                    "@type": "Alarm",
                    "externalAlarmId": "5551212",
                    "state": "raised",
                    "alarmType": "environmentalAlarm",
                    "perceivedSeverity": "minor",
                    "probableCause": "rectifierLowVoltage",
                    "specificProblem": "ps=3,sl=1,in=8",
                    "alarmedObjectType": "Rectifier",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "93051825",
                        "href": "/resourceInventoryManagement/v5/resource/93051825",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmDetails": "voltage=204",
                    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
                    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
```

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```
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                },
                {
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/8675399",
                    "@type": "Alarm",
                    "externalAlarmId": "5551234",
                    "state": "raised",
                    "alarmType": "equipmentAlarm",
                    "perceivedSeverity": "minor",
                    "probableCause": "powerSupplyFailure",
                    "specificProblem": "ch=1,ps=3",
                    "alarmedObjectType": "Chassis",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "95830731",
                        "href": "/resourceInventoryManagement/v5/resource/95830731",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
                    "alarmReportingTime": "2019-07-03T03:32:18.102Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "815",
        "name": "APP-663",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "103",
        "name": "APP-57",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@type": "UnGroupAlarmCreateEvent"
}
```


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### Delete Event

Message example for AlarmDeleteEvent event

```
Content-Type: application/json

{
    "correlationId": "a15a9002-cd5f",
    "description": "AlarmDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "4e92-b13d-4e9a81761f13",
    "eventTime": "2022-08-25T12:20:19.557Z",
    "eventType": "AlarmDeleteEvent",
    "priority": "3",
    "timeOccurred": "2022-08-25T12:20:17.467Z",
    "title": "AlarmDeleteEvent",
    "event": {
        "alarm": {
            "id": "8675309",
            "href": "http://servername/alarm/8675309",
            "@type": "Alarm"
        }
    },
    "reportingSystem": {
        "id": "815",
        "name": "APP-663",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "103",
        "name": "APP-57",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@type": "AlarmDeleteEvent"
}
```

### Attribute Value Change Event

Message example for AlarmAttributeValueChangeEvent event

```
Content-Type: application/json

{
    "correlationId": "bc449e2c-34c0",
    "description": "AlarmAttributeValueChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "441f-9380-4858c53114c2",
    "eventTime": "2022-08-25T12:20:19.535Z",
    "eventType": "AlarmAttributeValueChangeEvent",
    "priority": "1",
    "timeOccurred": "2022-08-25T12:20:19.184Z",
    "title": "AlarmAttributeValueChangeEvent",
    "event": {
        "alarm": {
            "href": "http://servername/alarm/8675309",
            "id": "8675309",
```

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```
            "@type": "Alarm",
            "comment": [
                {
                    "@type": "Comment",
                    "userId": "bob@example.net",
                    "systemId": "ems-1",
                    "time": "2019-07-03T03:37:33.827Z",
                    "comment": "Dispatched"
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "815",
        "name": "APP-663",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "103",
        "name": "APP-57",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "AlarmAttributeValueChangeEvent"
}
```


### State Change Event

Message example for AlarmStateChangeEvent event


```
Content-Type: application/json

{
    "correlationId": "2bb14d44-8dd6",
    "description": "UnGroupAlarmStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "4d5c-a34c-38218ba1fd66",
    "eventTime": "2022-08-25T12:20:19.710Z",
    "eventType": "UnGroupAlarmStateChangeEvent",
    "priority": "3",
    "timeOccurred": "2022-08-25T12:20:15.920Z",
    "title": "UnGroupAlarmStateChangeEvent",
    "event": {
        "unGroupAlarm": {
            "@type": "UnGroupAlarm",
            "id": "1069",
            "href": "/alarmManagement/v5/unGroupAlarm/1069",
            "sourceSystemId": "ems-1",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "parentAlarm": {
                "@type": "Alarm",
                "id": "8675309"
            },
            "correlatedAlarm": [
                {
                    "@type": "Alarm",
                    "id": "8675399"
                },
```


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```
                {
                    "@type": "Alarm",
                    "id": "8675399"
                }
            ],
            "state": "done",
            "unGroupedAlarm": [
                {
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/8675309",
                    "@type": "Alarm",
                    "externalAlarmId": "5551212",
                    "state": "updated",
                    "alarmType": "environmentalAlarm",
                    "perceivedSeverity": "major",
                    "probableCause": "rectifierLowVoltage",
                    "specificProblem": "ps=3,sl=1,in=8",
                    "alarmedObjectType": "Rectifier",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "93051825",
                        "href": "/resourceInventoryManagement/v5/resource/93051825",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmDetails": "voltage=95",
                    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
                    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                },
                {
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/8675399",
                    "@type": "Alarm",
                    "externalAlarmId": "5551234",
                    "state": "updated",
                    "alarmType": "equipmentAlarm",
                    "perceivedSeverity": "minor",
                    "probableCause": "powerSupplyFailure",
                    "specificProblem": "ch=1,ps=3",
                    "alarmedObjectType": "Chassis",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "95830731",
                        "href": "/resourceInventoryManagement/v5/resource/95830731",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
                    "alarmReportingTime": "2019-07-03T03:32:18.102Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
```


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```
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "815",
        "name": "APP-663",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "103",
        "name": "APP-57",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "UnGroupAlarmStateChangeEvent"
}
```


## AckAlarm

### Create Event

Message example for AckAlarmCreateEvent event


```
Content-Type: application/json

{
    "correlationId": "96b37351-efde",
    "description": "UnGroupAlarmCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4505-85a4-b1a0c5587ebc",
    "eventTime": "2022-08-25T12:20:19.700Z",
    "eventType": "UnGroupAlarmCreateEvent",
    "priority": "3",
    "timeOccurred": "2022-08-25T12:20:10.499Z",
    "title": "UnGroupCreateEvent",
    "event": {
        "unGroupAlarm": {
            "@type": "UnGroupAlarm",
            "id": "1069",
            "href": "http://servername/unGroupAlarm/1069",
            "sourceSystemId": "ems-1",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "parentAlarm": {
                "@type": "Alarm",
```


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```
            "id": "8675309"
        },
        "correlatedAlarm": [
            {
                "@type": "Alarm",
                "id": "8675399"
            },
            {
                "@type": "Alarm",
                "id": "8675399"
            }
        ],
        "state": "done",
        "unGroupedAlarm": [
            {
                "id": "8675309",
                "href": "/alarmManagement/v5/alarm/8675309",
                "@type": "Alarm",
                "externalAlarmId": "5551212",
                "state": "raised",
                "alarmType": "environmentalAlarm",
                "perceivedSeverity": "minor",
                "probableCause": "rectifierLowVoltage",
                "specificProblem": "ps=3,sl=1,in=8",
                "alarmedObjectType": "Rectifier",
                "alarmedObject": {
                    "@type": "AlarmedObjectRef",
                    "id": "93051825",
                    "href": "/resourceInventoryManagement/v5/resource/93051825",
                    "@referredType": "Resource"
                },
                "sourceSystemId": "ems-1",
                "alarmDetails": "voltage=204",
                "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
                "alarmReportingTime": "2019-07-03T03:32:17.552Z",
                "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                "ackSystemId": "ems-1",
                "ackUserId": "bob@example.net",
                "ackState": "unacknowledged",
                "comment": [
                    {
                        "@type": "Comment",
                        "userId": "bob@example.net",
                        "systemId": "ems-1",
                        "time": "2019-07-03T03:37:33.827Z",
                        "comment": "Dispatched"
                    }
                ]
            },
            {
                "id": "8675399",
                "href": "/alarmManagement/v5/alarm/8675399",
                "@type": "Alarm",
                "externalAlarmId": "5551234",
                "state": "raised",
                "alarmType": "equipmentAlarm",
                "perceivedSeverity": "minor",
                "probableCause": "powerSupplyFailure",
                "specificProblem": "ch=1,ps=3",
                "alarmedObjectType": "Chassis",
                "alarmedObject": {
                    "@type": "AlarmedObjectRef",
                    "id": "95830731",
```


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```
                        "href": "/resourceInventoryManagement/v5/resource/95830731",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
                    "alarmReportingTime": "2019-07-03T03:32:18.102Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "815",
        "name": "APP-663",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "103",
        "name": "APP-57",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@type": "UnGroupAlarmCreateEvent"
}
```


### State Change Event

Message example for AckAlarmStateChangeEvent event


```
Content-Type: application/json

{
    "correlationId": "2bb14d44-8dd6",
    "description": "UnGroupAlarmStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "4d5c-a34c-38218ba1fd66",
    "eventTime": "2022-08-25T12:20:19.710Z",
    "eventType": "UnGroupAlarmStateChangeEvent",
    "priority": "3",
    "timeOccurred": "2022-08-25T12:20:15.920Z",
    "title": "UnGroupAlarmStateChangeEvent",
    "event": {
        "unGroupAlarm": {
            "@type": "UnGroupAlarm",
            "id": "1069",
            "href": "/alarmManagement/v5/unGroupAlarm/1069",
            "sourceSystemId": "ems-1",
```


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```
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "parentAlarm": {
                "@type": "Alarm",
                "id": "8675309"
            },
            "correlatedAlarm": [
                {
                    "@type": "Alarm",
                    "id": "8675399"
                },
                {
                    "@type": "Alarm",
                    "id": "8675399"
                }
            ],
            "state": "done",
            "unGroupedAlarm": [
                {
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/8675309",
                    "@type": "Alarm",
                    "externalAlarmId": "5551212",
                    "state": "updated",
                    "alarmType": "environmentalAlarm",
                    "perceivedSeverity": "major",
                    "probableCause": "rectifierLowVoltage",
                    "specificProblem": "ps=3,sl=1,in=8",
                    "alarmedObjectType": "Rectifier",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "93051825",
                        "href": "/resourceInventoryManagement/v5/resource/93051825",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmDetails": "voltage=95",
                    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
                    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                },
                {
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/8675399",
                    "@type": "Alarm",
                    "externalAlarmId": "5551234",
                    "state": "updated",
                    "alarmType": "equipmentAlarm",
                    "perceivedSeverity": "minor",
                    "probableCause": "powerSupplyFailure",
                    "specificProblem": "ch=1,ps=3",
                    "alarmedObjectType": "Chassis",
```


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```
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "95830731",
                        "href": "/resourceInventoryManagement/v5/resource/95830731",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
                    "alarmReportingTime": "2019-07-03T03:32:18.102Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "815",
        "name": "APP-663",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "103",
        "name": "APP-57",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "UnGroupAlarmStateChangeEvent"
}
```


## UnAckAlarm

### Create Event

Message example for UnAckAlarmCreateEvent event


```
Content-Type: application/json

{
    "correlationId": "96b37351-efde",
    "description": "UnGroupAlarmCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4505-85a4-b1a0c5587ebc",
    "eventTime": "2022-08-25T12:20:19.700Z",
    "eventType": "UnGroupAlarmCreateEvent",
    "priority": "3",
    "timeOccurred": "2022-08-25T12:20:10.499Z",
```


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```
    "title": "UnGroupCreateEvent",
    "event": {
        "unGroupAlarm": {
            "@type": "UnGroupAlarm",
            "id": "1069",
            "href": "http://servername/unGroupAlarm/1069",
            "sourceSystemId": "ems-1",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "parentAlarm": {
                "@type": "Alarm",
                "id": "8675309"
            },
            "correlatedAlarm": [
                {
                    "@type": "Alarm",
                    "id": "8675399"
                },
                {
                    "@type": "Alarm",
                    "id": "8675399"
                }
            ],
            "state": "done",
            "unGroupedAlarm": [
                {
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/8675309",
                    "@type": "Alarm",
                    "externalAlarmId": "5551212",
                    "state": "raised",
                    "alarmType": "environmentalAlarm",
                    "perceivedSeverity": "minor",
                    "probableCause": "rectifierLowVoltage",
                    "specificProblem": "ps=3,sl=1,in=8",
                    "alarmedObjectType": "Rectifier",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "93051825",
                        "href": "/resourceInventoryManagement/v5/resource/93051825",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmDetails": "voltage=204",
                    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
                    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                },
                {
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/8675399",
                    "@type": "Alarm",
```


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```
                    "externalAlarmId": "5551234",
                    "state": "raised",
                    "alarmType": "equipmentAlarm",
                    "perceivedSeverity": "minor",
                    "probableCause": "powerSupplyFailure",
                    "specificProblem": "ch=1,ps=3",
                    "alarmedObjectType": "Chassis",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "95830731",
                        "href": "/resourceInventoryManagement/v5/resource/95830731",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
                    "alarmReportingTime": "2019-07-03T03:32:18.102Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "815",
        "name": "APP-663",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "103",
        "name": "APP-57",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@type": "UnGroupAlarmCreateEvent"
}
```


### State Change Event

Message example for UnAckAlarmStateChangeEvent event


```
Content-Type: application/json

{
    "correlationId": "2bb14d44-8dd6",
    "description": "UnGroupAlarmStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "4d5c-a34c-38218ba1fd66",
    "eventTime": "2022-08-25T12:20:19.710Z",
```


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```
    "eventType": "UnGroupAlarmStateChangeEvent",
    "priority": "3",
    "timeOccurred": "2022-08-25T12:20:15.920Z",
    "title": "UnGroupAlarmStateChangeEvent",
    "event": {
        "unGroupAlarm": {
            "@type": "UnGroupAlarm",
            "id": "1069",
            "href": "/alarmManagement/v5/unGroupAlarm/1069",
            "sourceSystemId": "ems-1",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "parentAlarm": {
                "@type": "Alarm",
                "id": "8675309"
            },
            "correlatedAlarm": [
                {
                    "@type": "Alarm",
                    "id": "8675399"
                },
                {
                    "@type": "Alarm",
                    "id": "8675399"
                }
            ],
            "state": "done",
            "unGroupedAlarm": [
                {
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/8675309",
                    "@type": "Alarm",
                    "externalAlarmId": "5551212",
                    "state": "updated",
                    "alarmType": "environmentalAlarm",
                    "perceivedSeverity": "major",
                    "probableCause": "rectifierLowVoltage",
                    "specificProblem": "ps=3,sl=1,in=8",
                    "alarmedObjectType": "Rectifier",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "93051825",
                        "href": "/resourceInventoryManagement/v5/resource/93051825",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmDetails": "voltage=95",
                    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
                    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                },
                {
```


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```
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/8675399",
                    "@type": "Alarm",
                    "externalAlarmId": "5551234",
                    "state": "updated",
                    "alarmType": "equipmentAlarm",
                    "perceivedSeverity": "minor",
                    "probableCause": "powerSupplyFailure",
                    "specificProblem": "ch=1,ps=3",
                    "alarmedObjectType": "Chassis",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "95830731",
                        "href": "/resourceInventoryManagement/v5/resource/95830731",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
                    "alarmReportingTime": "2019-07-03T03:32:18.102Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "815",
        "name": "APP-663",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "103",
        "name": "APP-57",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "UnGroupAlarmStateChangeEvent"
}
```


## ClearAlarm

### Create Event

Message example for ClearAlarmCreateEvent event


```
Content-Type: application/json
```


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```
{
    "correlationId": "96b37351-efde",
    "description": "UnGroupAlarmCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4505-85a4-b1a0c5587ebc",
    "eventTime": "2022-08-25T12:20:19.700Z",
    "eventType": "UnGroupAlarmCreateEvent",
    "priority": "3",
    "timeOccurred": "2022-08-25T12:20:10.499Z",
    "title": "UnGroupCreateEvent",
    "event": {
        "unGroupAlarm": {
            "@type": "UnGroupAlarm",
            "id": "1069",
            "href": "http://servername/unGroupAlarm/1069",
            "sourceSystemId": "ems-1",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "parentAlarm": {
                "@type": "Alarm",
                "id": "8675309"
            },
            "correlatedAlarm": [
                {
                    "@type": "Alarm",
                    "id": "8675399"
                },
                {
                    "@type": "Alarm",
                    "id": "8675399"
                }
            ],
            "state": "done",
            "unGroupedAlarm": [
                {
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/8675309",
                    "@type": "Alarm",
                    "externalAlarmId": "5551212",
                    "state": "raised",
                    "alarmType": "environmentalAlarm",
                    "perceivedSeverity": "minor",
                    "probableCause": "rectifierLowVoltage",
                    "specificProblem": "ps=3,sl=1,in=8",
                    "alarmedObjectType": "Rectifier",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "93051825",
                        "href": "/resourceInventoryManagement/v5/resource/93051825",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmDetails": "voltage=204",
                    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
                    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
```


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```
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                },
                {
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/8675399",
                    "@type": "Alarm",
                    "externalAlarmId": "5551234",
                    "state": "raised",
                    "alarmType": "equipmentAlarm",
                    "perceivedSeverity": "minor",
                    "probableCause": "powerSupplyFailure",
                    "specificProblem": "ch=1,ps=3",
                    "alarmedObjectType": "Chassis",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "95830731",
                        "href": "/resourceInventoryManagement/v5/resource/95830731",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
                    "alarmReportingTime": "2019-07-03T03:32:18.102Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "815",
        "name": "APP-663",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "103",
        "name": "APP-57",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@type": "UnGroupAlarmCreateEvent"
}
```


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### State Change Event

Message example for ClearAlarmStateChangeEvent event

```
Content-Type: application/json

{
    "correlationId": "2bb14d44-8dd6",
    "description": "UnGroupAlarmStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "4d5c-a34c-38218ba1fd66",
    "eventTime": "2022-08-25T12:20:19.710Z",
    "eventType": "UnGroupAlarmStateChangeEvent",
    "priority": "3",
    "timeOccurred": "2022-08-25T12:20:15.920Z",
    "title": "UnGroupAlarmStateChangeEvent",
    "event": {
        "unGroupAlarm": {
            "@type": "UnGroupAlarm",
            "id": "1069",
            "href": "/alarmManagement/v5/unGroupAlarm/1069",
            "sourceSystemId": "ems-1",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "parentAlarm": {
                "@type": "Alarm",
                "id": "8675309"
            },
            "correlatedAlarm": [
                {
                    "@type": "Alarm",
                    "id": "8675399"
                },
                {
                    "@type": "Alarm",
                    "id": "8675399"
                }
            ],
            "state": "done",
            "unGroupedAlarm": [
                {
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/8675309",
                    "@type": "Alarm",
                    "externalAlarmId": "5551212",
                    "state": "updated",
                    "alarmType": "environmentalAlarm",
                    "perceivedSeverity": "major",
                    "probableCause": "rectifierLowVoltage",
                    "specificProblem": "ps=3,sl=1,in=8",
                    "alarmedObjectType": "Rectifier",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "93051825",
                        "href": "/resourceInventoryManagement/v5/resource/93051825",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmDetails": "voltage=95",
                    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
                    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
```

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```
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                },
                {
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/8675399",
                    "@type": "Alarm",
                    "externalAlarmId": "5551234",
                    "state": "updated",
                    "alarmType": "equipmentAlarm",
                    "perceivedSeverity": "minor",
                    "probableCause": "powerSupplyFailure",
                    "specificProblem": "ch=1,ps=3",
                    "alarmedObjectType": "Chassis",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "95830731",
                        "href": "/resourceInventoryManagement/v5/resource/95830731",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
                    "alarmReportingTime": "2019-07-03T03:32:18.102Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "815",
        "name": "APP-663",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "103",
        "name": "APP-57",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
```


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```
    "@type": "UnGroupAlarmStateChangeEvent"
}
```

## CommentAlarm

### Create Event

Message example for CommentAlarmCreateEvent event

```
Content-Type: application/json

{
    "correlationId": "96b37351-efde",
    "description": "UnGroupAlarmCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4505-85a4-b1a0c5587ebc",
    "eventTime": "2022-08-25T12:20:19.700Z",
    "eventType": "UnGroupAlarmCreateEvent",
    "priority": "3",
    "timeOccurred": "2022-08-25T12:20:10.499Z",
    "title": "UnGroupCreateEvent",
    "event": {
        "unGroupAlarm": {
            "@type": "UnGroupAlarm",
            "id": "1069",
            "href": "http://servername/unGroupAlarm/1069",
            "sourceSystemId": "ems-1",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "parentAlarm": {
                "@type": "Alarm",
                "id": "8675309"
            },
            "correlatedAlarm": [
                {
                    "@type": "Alarm",
                    "id": "8675399"
                },
                {
                    "@type": "Alarm",
                    "id": "8675399"
                }
            ],
            "state": "done",
            "unGroupedAlarm": [
                {
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/8675309",
                    "@type": "Alarm",
                    "externalAlarmId": "5551212",
                    "state": "raised",
                    "alarmType": "environmentalAlarm",
                    "perceivedSeverity": "minor",
                    "probableCause": "rectifierLowVoltage",
                    "specificProblem": "ps=3,sl=1,in=8",
                    "alarmedObjectType": "Rectifier",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "93051825",
                        "href": "/resourceInventoryManagement/v5/resource/93051825",
```

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```
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmDetails": "voltage=204",
                    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
                    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                },
                {
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/8675399",
                    "@type": "Alarm",
                    "externalAlarmId": "5551234",
                    "state": "raised",
                    "alarmType": "equipmentAlarm",
                    "perceivedSeverity": "minor",
                    "probableCause": "powerSupplyFailure",
                    "specificProblem": "ch=1,ps=3",
                    "alarmedObjectType": "Chassis",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "95830731",
                        "href": "/resourceInventoryManagement/v5/resource/95830731",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
                    "alarmReportingTime": "2019-07-03T03:32:18.102Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "815",
        "name": "APP-663",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
```


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```
    "source": {
        "id": "103",
        "name": "APP-57",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@type": "UnGroupAlarmCreateEvent"
}
```

### State Change Event

Message example for CommentAlarmStateChangeEvent event

```
Content-Type: application/json

{
    "correlationId": "2bb14d44-8dd6",
    "description": "UnGroupAlarmStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "4d5c-a34c-38218ba1fd66",
    "eventTime": "2022-08-25T12:20:19.710Z",
    "eventType": "UnGroupAlarmStateChangeEvent",
    "priority": "3",
    "timeOccurred": "2022-08-25T12:20:15.920Z",
    "title": "UnGroupAlarmStateChangeEvent",
    "event": {
        "unGroupAlarm": {
            "@type": "UnGroupAlarm",
            "id": "1069",
            "href": "/alarmManagement/v5/unGroupAlarm/1069",
            "sourceSystemId": "ems-1",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "parentAlarm": {
                "@type": "Alarm",
                "id": "8675309"
            },
            "correlatedAlarm": [
                {
                    "@type": "Alarm",
                    "id": "8675399"
                },
                {
                    "@type": "Alarm",
                    "id": "8675399"
                }
            ],
            "state": "done",
            "unGroupedAlarm": [
                {
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/8675309",
                    "@type": "Alarm",
                    "externalAlarmId": "5551212",
                    "state": "updated",
                    "alarmType": "environmentalAlarm",
                    "perceivedSeverity": "major",
                    "probableCause": "rectifierLowVoltage",
                    "specificProblem": "ps=3,sl=1,in=8",
                    "alarmedObjectType": "Rectifier",
                    "alarmedObject": {
```

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```
                        "@type": "AlarmedObjectRef",
                        "id": "93051825",
                        "href": "/resourceInventoryManagement/v5/resource/93051825",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmDetails": "voltage=95",
                    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
                    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                },
                {
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/8675399",
                    "@type": "Alarm",
                    "externalAlarmId": "5551234",
                    "state": "updated",
                    "alarmType": "equipmentAlarm",
                    "perceivedSeverity": "minor",
                    "probableCause": "powerSupplyFailure",
                    "specificProblem": "ch=1,ps=3",
                    "alarmedObjectType": "Chassis",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "95830731",
                        "href": "/resourceInventoryManagement/v5/resource/95830731",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
                    "alarmReportingTime": "2019-07-03T03:32:18.102Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "815",
        "name": "APP-663",
```


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```
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "103",
        "name": "APP-57",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "UnGroupAlarmStateChangeEvent"
}
```


## GroupAlarm

### Create Event

Message example for GroupAlarmCreateEvent event


```
Content-Type: application/json

{
    "correlationId": "96b37351-efde",
    "description": "UnGroupAlarmCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4505-85a4-b1a0c5587ebc",
    "eventTime": "2022-08-25T12:20:19.700Z",
    "eventType": "UnGroupAlarmCreateEvent",
    "priority": "3",
    "timeOccurred": "2022-08-25T12:20:10.499Z",
    "title": "UnGroupCreateEvent",
    "event": {
        "unGroupAlarm": {
            "@type": "UnGroupAlarm",
            "id": "1069",
            "href": "http://servername/unGroupAlarm/1069",
            "sourceSystemId": "ems-1",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "parentAlarm": {
                "@type": "Alarm",
                "id": "8675309"
            },
            "correlatedAlarm": [
                {
                    "@type": "Alarm",
                    "id": "8675399"
                },
                {
                    "@type": "Alarm",
                    "id": "8675399"
                }
            ],
            "state": "done",
            "unGroupedAlarm": [
                {
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/8675309",
                    "@type": "Alarm",
                    "externalAlarmId": "5551212",
```


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```
                    "state": "raised",
                    "alarmType": "environmentalAlarm",
                    "perceivedSeverity": "minor",
                    "probableCause": "rectifierLowVoltage",
                    "specificProblem": "ps=3,sl=1,in=8",
                    "alarmedObjectType": "Rectifier",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "93051825",
                        "href": "/resourceInventoryManagement/v5/resource/93051825",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmDetails": "voltage=204",
                    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
                    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                },
                {
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/8675399",
                    "@type": "Alarm",
                    "externalAlarmId": "5551234",
                    "state": "raised",
                    "alarmType": "equipmentAlarm",
                    "perceivedSeverity": "minor",
                    "probableCause": "powerSupplyFailure",
                    "specificProblem": "ch=1,ps=3",
                    "alarmedObjectType": "Chassis",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "95830731",
                        "href": "/resourceInventoryManagement/v5/resource/95830731",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
                    "alarmReportingTime": "2019-07-03T03:32:18.102Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
```


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```
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "815",
        "name": "APP-663",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "103",
        "name": "APP-57",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@type": "UnGroupAlarmCreateEvent"
}
```

### State Change Event

Message example for GroupAlarmStateChangeEvent event

```
Content-Type: application/json

{
    "correlationId": "2bb14d44-8dd6",
    "description": "UnGroupAlarmStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "4d5c-a34c-38218ba1fd66",
    "eventTime": "2022-08-25T12:20:19.710Z",
    "eventType": "UnGroupAlarmStateChangeEvent",
    "priority": "3",
    "timeOccurred": "2022-08-25T12:20:15.920Z",
    "title": "UnGroupAlarmStateChangeEvent",
    "event": {
        "unGroupAlarm": {
            "@type": "UnGroupAlarm",
            "id": "1069",
            "href": "/alarmManagement/v5/unGroupAlarm/1069",
            "sourceSystemId": "ems-1",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "parentAlarm": {
                "@type": "Alarm",
                "id": "8675309"
            },
            "correlatedAlarm": [
                {
                    "@type": "Alarm",
                    "id": "8675399"
                },
                {
                    "@type": "Alarm",
                    "id": "8675399"
                }
            ],
            "state": "done",
            "unGroupedAlarm": [
                {
                    "id": "8675309",
```

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```
                "href": "/alarmManagement/v5/alarm/8675309",
                "@type": "Alarm",
                "externalAlarmId": "5551212",
                "state": "updated",
                "alarmType": "environmentalAlarm",
                "perceivedSeverity": "major",
                "probableCause": "rectifierLowVoltage",
                "specificProblem": "ps=3,sl=1,in=8",
                "alarmedObjectType": "Rectifier",
                "alarmedObject": {
                    "@type": "AlarmedObjectRef",
                    "id": "93051825",
                    "href": "/resourceInventoryManagement/v5/resource/93051825",
                    "@referredType": "Resource"
                },
                "sourceSystemId": "ems-1",
                "alarmDetails": "voltage=95",
                "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
                "alarmReportingTime": "2019-07-03T03:32:17.552Z",
                "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                "ackSystemId": "ems-1",
                "ackUserId": "bob@example.net",
                "ackState": "unacknowledged",
                "comment": [
                    {
                        "@type": "Comment",
                        "userId": "bob@example.net",
                        "systemId": "ems-1",
                        "time": "2019-07-03T03:37:33.827Z",
                        "comment": "Dispatched"
                    }
                ]
            },
            {
                "id": "8675399",
                "href": "/alarmManagement/v5/alarm/8675399",
                "@type": "Alarm",
                "externalAlarmId": "5551234",
                "state": "updated",
                "alarmType": "equipmentAlarm",
                "perceivedSeverity": "minor",
                "probableCause": "powerSupplyFailure",
                "specificProblem": "ch=1,ps=3",
                "alarmedObjectType": "Chassis",
                "alarmedObject": {
                    "@type": "AlarmedObjectRef",
                    "id": "95830731",
                    "href": "/resourceInventoryManagement/v5/resource/95830731",
                    "@referredType": "Resource"
                },
                "sourceSystemId": "ems-1",
                "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
                "alarmReportingTime": "2019-07-03T03:32:18.102Z",
                "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                "ackSystemId": "ems-1",
                "ackUserId": "bob@example.net",
                "ackState": "unacknowledged",
                "comment": [
                    {
                        "@type": "Comment",
                        "userId": "bob@example.net",
                        "systemId": "ems-1",
                        "time": "2019-07-03T03:37:33.827Z",
```


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```
                            "comment": "Dispatched"
                        }
                    ]
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "815",
        "name": "APP-663",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "103",
        "name": "APP-57",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "UnGroupAlarmStateChangeEvent"
}
```


## UnGroupAlarm

### Create Event

Message example for UnGroupAlarmCreateEvent event


```
Content-Type: application/json

{
    "correlationId": "96b37351-efde",
    "description": "UnGroupAlarmCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4505-85a4-b1a0c5587ebc",
    "eventTime": "2022-08-25T12:20:19.700Z",
    "eventType": "UnGroupAlarmCreateEvent",
    "priority": "3",
    "timeOccurred": "2022-08-25T12:20:10.499Z",
    "title": "UnGroupCreateEvent",
    "event": {
        "unGroupAlarm": {
            "@type": "UnGroupAlarm",
            "id": "1069",
            "href": "http://servername/unGroupAlarm/1069",
            "sourceSystemId": "ems-1",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "parentAlarm": {
                "@type": "Alarm",
                "id": "8675309"
            },
            "correlatedAlarm": [
                {
                    "@type": "Alarm",
                    "id": "8675399"
                },
                {
                    "@type": "Alarm",
```


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```
                    "id": "8675399"
                }
            ],
            "state": "done",
            "unGroupedAlarm": [
                {
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/8675309",
                    "@type": "Alarm",
                    "externalAlarmId": "5551212",
                    "state": "raised",
                    "alarmType": "environmentalAlarm",
                    "perceivedSeverity": "minor",
                    "probableCause": "rectifierLowVoltage",
                    "specificProblem": "ps=3,sl=1,in=8",
                    "alarmedObjectType": "Rectifier",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "93051825",
                        "href": "/resourceInventoryManagement/v5/resource/93051825",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmDetails": "voltage=204",
                    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
                    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                },
                {
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/8675399",
                    "@type": "Alarm",
                    "externalAlarmId": "5551234",
                    "state": "raised",
                    "alarmType": "equipmentAlarm",
                    "perceivedSeverity": "minor",
                    "probableCause": "powerSupplyFailure",
                    "specificProblem": "ch=1,ps=3",
                    "alarmedObjectType": "Chassis",
                    "alarmedObject": {
                        "@type": "AlarmedObjectRef",
                        "id": "95830731",
                        "href": "/resourceInventoryManagement/v5/resource/95830731",
                        "@referredType": "Resource"
                    },
                    "sourceSystemId": "ems-1",
                    "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
                    "alarmReportingTime": "2019-07-03T03:32:18.102Z",
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
```


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```
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "815",
        "name": "APP-663",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "103",
        "name": "APP-57",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@type": "UnGroupAlarmCreateEvent"
}
```


### State Change Event

Message example for UnGroupAlarmStateChangeEvent event


```
Content-Type: application/json

{
    "correlationId": "2bb14d44-8dd6",
    "description": "UnGroupAlarmStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "4d5c-a34c-38218ba1fd66",
    "eventTime": "2022-08-25T12:20:19.710Z",
    "eventType": "UnGroupAlarmStateChangeEvent",
    "priority": "3",
    "timeOccurred": "2022-08-25T12:20:15.920Z",
    "title": "UnGroupAlarmStateChangeEvent",
    "event": {
        "unGroupAlarm": {
            "@type": "UnGroupAlarm",
            "id": "1069",
            "href": "/alarmManagement/v5/unGroupAlarm/1069",
            "sourceSystemId": "ems-1",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "parentAlarm": {
                "@type": "Alarm",
                "id": "8675309"
            },
            "correlatedAlarm": [
                {
                    "@type": "Alarm",
                    "id": "8675399"
```


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```
            },
            {
                "@type": "Alarm",
                "id": "8675399"
            }
        ],
        "state": "done",
        "unGroupedAlarm": [
            {
                "id": "8675309",
                "href": "/alarmManagement/v5/alarm/8675309",
                "@type": "Alarm",
                "externalAlarmId": "5551212",
                "state": "updated",
                "alarmType": "environmentalAlarm",
                "perceivedSeverity": "major",
                "probableCause": "rectifierLowVoltage",
                "specificProblem": "ps=3,sl=1,in=8",
                "alarmedObjectType": "Rectifier",
                "alarmedObject": {
                    "@type": "AlarmedObjectRef",
                    "id": "93051825",
                    "href": "/resourceInventoryManagement/v5/resource/93051825",
                    "@referredType": "Resource"
                },
                "sourceSystemId": "ems-1",
                "alarmDetails": "voltage=95",
                "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
                "alarmReportingTime": "2019-07-03T03:32:17.552Z",
                "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                "ackSystemId": "ems-1",
                "ackUserId": "bob@example.net",
                "ackState": "unacknowledged",
                "comment": [
                    {
                        "@type": "Comment",
                        "userId": "bob@example.net",
                        "systemId": "ems-1",
                        "time": "2019-07-03T03:37:33.827Z",
                        "comment": "Dispatched"
                    }
                ]
            },
            {
                "id": "8675399",
                "href": "/alarmManagement/v5/alarm/8675399",
                "@type": "Alarm",
                "externalAlarmId": "5551234",
                "state": "updated",
                "alarmType": "equipmentAlarm",
                "perceivedSeverity": "minor",
                "probableCause": "powerSupplyFailure",
                "specificProblem": "ch=1,ps=3",
                "alarmedObjectType": "Chassis",
                "alarmedObject": {
                    "@type": "AlarmedObjectRef",
                    "id": "95830731",
                    "href": "/resourceInventoryManagement/v5/resource/95830731",
                    "@referredType": "Resource"
                },
                "sourceSystemId": "ems-1",
                "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
                "alarmReportingTime": "2019-07-03T03:32:18.102Z",
```


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```
                    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                    "ackSystemId": "ems-1",
                    "ackUserId": "bob@example.net",
                    "ackState": "unacknowledged",
                    "comment": [
                        {
                            "@type": "Comment",
                            "userId": "bob@example.net",
                            "systemId": "ems-1",
                            "time": "2019-07-03T03:37:33.827Z",
                            "comment": "Dispatched"
                        }
                    ]
                }
            ]
        }
    },
    "reportingSystem": {
        "id": "815",
        "name": "APP-663",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "103",
        "name": "APP-57",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "UnGroupAlarmStateChangeEvent"
}
```


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# API OPERATIONS

Remember the following Uniform Contract:

| Operation on Entities | Uniform API Operation | Description |
|---|---|---|
| Query Entities | GET Resource | GET must be used to retrieve a representation of a resource. |
| Create Entity | POST Resource | POST must be used to create a new resource |
| Partial Update of an Entity | PATCH Resource | PATCH must be used to partially update a resource |
| Remove an Entity | DELETE Resource | DELETE must be used to remove a resource |
| Execute an Action on an Entity | POST on TASK Resource | POST must be used to execute Task Resources |
| Other Request Methods | POST on TASK Resource | GET and POST must not be used to tunnel other request methods. |

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

## Operations on Alarm

### Retrieves a Alarm by ID

**`GET /alarm/{id}?fields=…`**

#### Description

This operation retrieves an Alarm entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

#### Usage samples

Example of a request for retrieving an alarm:

#### Request

```
GET /alarm/8675309
Content-Type: application/json
```

#### Response

```
200
```

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```
{
    "id": "8675309",
    "href": "/alarmManagement/v5/alarm/8675309",
    "@type": "Alarm",
    "externalAlarmId": "5551212",
    "state": "updated",
    "alarmType": "environmentalAlarm",
    "perceivedSeverity": "major",
    "probableCause": "rectifierLowVoltage",
    "specificProblem": "ps=3,sl=1,in=8",
    "alarmedObjectType": "Rectifier",
    "alarmedObject": {
        "@type": "AlarmedObjectRef",
        "@referredType": "Resource",
        "id": "93051825",
        "href": "/resourceInventoryManagement/v5/resource/93051825"
    },
    "sourceSystemId": "ems-1",
    "alarmDetails": "voltage=95",
    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
    "alarmChangedTime": "2019-07-03T03:32:24.715Z",
    "ackState": "unacknowledged"
}
```


## List or find Alarm objects

**`GET /alarm?fields=…`**

### Description

This operation list Alarm entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

### Usage samples

Example of a request for retrieving a list of alarms:

### Request

```
GET /alarm
Content-Type: application/json
```

### Response


```
200

[
    {
        "id": "8675309",
        "href": "/alarmManagement/v5/alarm/8675309",
        "@type": "Alarm",
        "externalAlarmId": "5551212",
        "state": "updated",
```


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```
        "alarmType": "environmentalAlarm",
        "perceivedSeverity": "major",
        "probableCause": "rectifierLowVoltage",
        "specificProblem": "ps=3,sl=1,in=8",
        "alarmedObjectType": "Rectifier",
        "alarmedObject": {
            "@type": "AlarmedObjectRef",
            "@referredType": "Resource",
            "id": "93051825",
            "href": "/resourceInventoryManagement/v5/resource/93051825"
        },
        "sourceSystemId": "ems-1",
        "alarmDetails": "voltage=95",
        "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
        "alarmReportingTime": "2019-07-03T03:32:17.552Z",
        "alarmChangedTime": "2019-07-03T03:32:24.715Z"
    },
    {
        "id": "8675399",
        "href": "/alarmManagement/v5/alarm/8675399",
        "@type": "Alarm",
        "externalAlarmId": "5551234",
        "state": "updated",
        "alarmType": "equipmentAlarm",
        "perceivedSeverity": "minor",
        "probableCause": "powerSupplyFailure",
        "specificProblem": "ch=1,ps=3",
        "alarmedObjectType": "Chassis",
        "alarmedObject": {
            "@type": "AlarmedObjectRef",
            "@referredType": "Resource",
            "id": "93051831",
            "href": "/resourceInventoryManagement/v5/resource/93051831"
        },
        "sourceSystemId": "ems-1",
        "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
        "alarmReportingTime": "2019-07-03T03:32:18.102Z"
    }
]
```


## Creates a Alarm

**`POST /alarm?fields=…`**

### Description

This operation creates an Alarm entity.

### Mandatory Attributes

| Mandatory Attributes | Rule |
|---|---|
| alarmRaisedTime | |
| alarmType | |
| alarmedObject | |
| alarmedObject.@type | |

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| Mandatory Attributes | Rule |
|---|---|
| perceivedSeverity | |
| probableCause | |
| sourceSystemId | |
| state | |
| @type | |

### Usage samples

Example of a request for creating an alarm:

### Request

```
POST /alarm
Content-Type: application/json

{
    "externalAlarmId": "5551212",
    "state": "raised",
    "@type": "Alarm",
    "alarmType": "environmentalAlarm",
    "perceivedSeverity": "minor",
    "probableCause": "rectifierLowVoltage",
    "specificProblem": "ps=3,sl=1,in=8",
    "alarmedObjectType": "NePowerSupply",
    "alarmedObject": {
        "@type": "AlarmedObjectRef",
        "@referredType": "Resource",
        "id": "93051825",
        "href": "/resourceInventoryManagement/v5/resource/93051825"
    },
    "sourceSystemId": "ems-1",
    "alarmDetails": "voltage=204",
    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
    "alarmReportingTime": "2019-07-03T03:32:17.552Z"
}
```

### Response

```
201

{
    "id": "8675309",
    "@type": "Alarm",
    "href": "/alarmManagement/v5/alarm/8675309",
    "externalAlarmId": "5551212",
    "state": "raised",
    "alarmType": "environmentalAlarm",
    "perceivedSeverity": "minor",
    "probableCause": "rectifierLowVoltage",
    "specificProblem": "ps=3,sl=1,in=8",
    "alarmedObjectType": "Rectifier",
    "alarmedObject": {
```

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```
        "@type": "AlarmedObjectRef",
        "@referredType": "Resource",
        "id": "93051825",
        "href": "/resourceInventoryManagement/v5/resource/93051825"
    },
    "sourceSystemId": "ems-1",
    "alarmDetails": "voltage=204",
    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
    "alarmReportingTime": "2019-07-03T03:32:17.552Z"
}
```

## Updates partially a Alarm

**`PATCH /alarm/{id}?fields=…`**

### Description

This operation allows partial updates of an Alarm entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

### Patchable and Non Patchable Attributes

| Non Patchable Attributes | Rule |
|---|---|
| alarmRaisedTime | |
| href | |
| id | |
| sourceSystemId | |
| @baseType | @baseType is immutable |
| @schemaLocation | @schemaLocation is immutable |
| @type | @type is immutable |

| Patchable Attributes | Rule |
|---|---|
| ackState | |
| ackSystemId | |
| ackUserId | |
| affectedService | |
| alarmChangedTime | |
| alarmClearedTime | |
| alarmDetails | |
| alarmEscalation | |

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| Patchable Attributes | Rule |
|---|---|
| alarmReportingTime | |
| alarmType | |
| alarmedObject | |
| alarmedObjectType | |
| clearSystemId | |
| clearUserId | |
| comment | |
| correlatedAlarm | |
| crossedThresholdInformation | |
| externalAlarmId | |
| isRootCause | |
| parentAlarm | |
| perceivedSeverity | |
| place | |
| plannedOutageIndicator | |
| probableCause | |
| proposedRepairedActions | |
| reportingSystemId | |
| serviceAffecting | |
| specificProblem | |
| state | |

### Usage samples

Example of updating an alarm using json-patch:

### Request

```
PATCH /alarm/8675309
Content-Type: application/json-patch+json

[
    {
        "op": "replace",
        "path": "/state",
        "value": "updated"
    },
    {
        "op": "replace",
        "path": "/perceivedSeverity",
        "value": "major"
    },
```

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```
    {
        "op": "replace",
        "path": "/alarmChangedTime",
        "value": "2019-07-03T03:32:24.715Z"
    },
    {
        "op": "replace",
        "path": "/alarmDetails",
        "value": "voltage=95"
    }
]
```

### Response

```
200

{
    "id": "8675309",
    "@type": "Alarm",
    "href": "/alarmManagement/v5/alarm/8675309",
    "externalAlarmId": "5551212",
    "state": "updated",
    "alarmType": "environmentalAlarm",
    "perceivedSeverity": "major",
    "probableCause": "rectifierLowVoltage",
    "specificProblem": "ps=3,sl=1,in=8",
    "alarmedObjectType": "Rectifier",
    "alarmedObject": {
        "@type": "AlarmedObjectRef",
        "@referredType": "Resource",
        "id": "93051825",
        "href": "/resourceInventoryManagement/v5/resource/93051825"
    },
    "sourceSystemId": "ems-1",
    "alarmDetails": "voltage=95",
    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
    "alarmChangedTime": "2019-07-03T03:32:24.715Z"
}
```

Example of updating an alarm:

### Request

```
PATCH /alarm/8675309
Content-Type: application/json

{
    "@type": "Alarm",
    "state": "updated",
    "perceivedSeverity": "major",
    "alarmDetails": "voltage=95",
    "alarmChangedTime": "2019-07-03T03:32:24.715Z"
}
```

### Response

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```
200

{
    "id": "8675309",
    "@type": "Alarm",
    "href": "/alarmManagement/v5/alarm/8675309",
    "externalAlarmId": "5551212",
    "state": "updated",
    "alarmType": "environmentalAlarm",
    "perceivedSeverity": "major",
    "probableCause": "rectifierLowVoltage",
    "specificProblem": "ps=3,sl=1,in=8",
    "alarmedObjectType": "Rectifier",
    "alarmedObject": {
        "@type": "AlarmedObjectRef",
        "@referredType": "Resource",
        "id": "93051825",
        "href": "/resourceInventoryManagement/v5/resource/93051825"
    },
    "sourceSystemId": "ems-1",
    "alarmDetails": "voltage=95",
    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
    "alarmChangedTime": "2019-07-03T03:32:24.715Z"
}
```

Example of updating an alarm using merge-patch:

### Request

```
PATCH /alarm/8675309
Content-Type: application/merge-patch+json

{
    "@type": "Alarm",
    "state": "updated",
    "perceivedSeverity": "major",
    "alarmDetails": "voltage=95",
    "alarmChangedTime": "2019-07-03T03:32:24.715Z"
}
```

### Response

```
200

{
    "id": "8675309",
    "@type": "Alarm",
    "href": "/alarmManagement/v5/alarm/8675309",
    "externalAlarmId": "5551212",
    "state": "updated",
    "alarmType": "environmentalAlarm",
    "perceivedSeverity": "major",
    "probableCause": "rectifierLowVoltage",
    "specificProblem": "ps=3,sl=1,in=8",
    "alarmedObjectType": "Rectifier",
```

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```
    "alarmedObject": {
        "@type": "AlarmedObjectRef",
        "@referredType": "Resource",
        "id": "93051825",
        "href": "/resourceInventoryManagement/v5/resource/93051825"
    },
    "sourceSystemId": "ems-1",
    "alarmDetails": "voltage=95",
    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
    "alarmChangedTime": "2019-07-03T03:32:24.715Z"
}
```

Example of acknowledging an alarm:

### Request

```
PATCH /alarm/8675309
Content-Type: application/json-patch+json

[
    {
        "op": "replace",
        "path": "/ackState",
        "value": "acknowledged"
    },
    {
        "op": "replace",
        "path": "/ackSystemId",
        "value": "ems-1"
    },
    {
        "op": "replace",
        "path": "/ackUserId",
        "value": "bob@example.net"
    },
    {
        "op": "replace",
        "path": "/ackTime",
        "value": "2019-07-03T03:33:12.623Z"
    }
]
```

### Response

```
200

{
    "id": "8675309",
    "@type": "Alarm",
    "href": "/alarmManagement/v5/alarm/8675309",
    "externalAlarmId": "5551212",
    "state": "updated",
    "alarmType": "environmentalAlarm",
    "perceivedSeverity": "major",
    "probableCause": "rectifierLowVoltage",
    "specificProblem": "ps=3,sl=1,in=8",
```

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```
    "alarmedObjectType": "Rectifier",
    "alarmedObject": {
        "@type": "AlarmedObjectRef",
        "@referredType": "Resource",
        "id": "93051825",
        "href": "/resourceInventoryManagement/v5/resource/93051825"
    },
    "sourceSystemId": "ems-1",
    "alarmDetails": "voltage=95",
    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
    "alarmChangedTime": "2019-07-03T03:32:24.715Z",
    "ackState": "acknowledged",
    "ackSystemId": "ems-1",
    "ackUserId": "bob@example.net"
}
```


Updating the alarm using json-patch-query+json:

### Request

```
PATCH /alarm/8675309
Content-Type: application/json-patch-query+json

[
    {
        "op": "replace",
        "path": "/parentAlarm/name?id=556698533215846321",
        "value": "Root cause"
    }
]
```

### Response


```
200

{
    "id": "8675309",
    "@type": "Alarm",
    "href": "/alarmManagement/v5/alarm/8675309",
    "externalAlarmId": "5551212",
    "state": "cleared",
    "alarmType": "environmentalAlarm",
    "perceivedSeverity": "cleared",
    "probableCause": "rectifierLowVoltage",
    "specificProblem": "ps=3,sl=1,in=8",
    "alarmedObjectType": "Rectifier",
    "alarmedObject": {
        "@type": "AlarmedObjectRef",
        "@referredType": "Resource",
        "id": "93051825",
        "href": "/resourceInventoryManagement/v5/resource/93051825"
    },
    "sourceSystemId": "ems-1",
    "alarmDetails": "voltage=95",
    "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
    "alarmReportingTime": "2019-07-03T03:32:17.552Z",
```


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```
    "alarmChangedTime": "2019-07-03T03:35:45.923Z",
    "alarmClearedTime": "2019-07-03T03:55:45.937Z",
    "clearSystemId": "ems-1",
    "clearUserId": "bob@example.net",
    "parentAlarm": [ {
        "@type": "AlarmRef",
        "id": "556698533215846321",
        "name": "Root cause"
    }

    ]
}
```


### Deletes a Alarm

**`DELETE /alarm/{id}`**

#### Description

This operation deletes an Alarm entity.

## Operations on AckAlarm

### Retrieves a AckAlarm by ID

**`GET /ackAlarm/{id}?fields=…`**

#### Description

This operation retrieves an AckAlarm entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

#### Usage samples

Example of a request for retrieving an acknowledge alarms task:

#### Request

```
GET /ackAlarm/1038
Content-Type: application/json
```

#### Response


```
200

{
    "id": "1038",
    "@type": "AckAlarm",
    "href": "/alarmManagement/v5/ackAlarm/1038",
    "ackSystemId": "ems-1",
    "ackUserId": "bob@example.net",
    "ackTime": "2019-07-03T03:33:12.623Z",
```


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```
    "alarmPattern": [
        {
            "@type": "Alarm",
            "id": "8675309"
        },
        {
            "@type": "Alarm",
            "id": "8675399"
        }
    ],
    "state": "inProgress",
    "ackedAlarm": []
}
```

## List or find AckAlarm objects

**`GET /ackAlarm?fields=…`**

### Description

This operation list AckAlarm entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

### Usage samples

Example of a request for retrieving a list of acknowledge alarms tasks:

### Request

```
GET /ackAlarm
Content-Type: application/json
```

### Response

```
200

[
    {
        "id": "1038",
        "@type": "AckAlarm",
        "href": "/alarmManagement/v5/ackAlarm/1038",
        "ackSystemId": "ems-1",
        "ackUserId": "bob@example.net",
        "ackTime": "2019-07-03T03:33:12.623Z",
        "alarmPattern": [
            {
                "@type": "Alarm",
                "id": "8675309"
            },
            {
                "@type": "Alarm",
                "id": "8675399"
            }
        ],
```

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```
        "state": "inProgress",
        "ackedAlarm": []
    }
]
```

## Creates a AckAlarm

### `POST /ackAlarm?fields=…`

#### Description

This operation creates an AckAlarm entity.

#### Mandatory Attributes

| Mandatory Attributes | Rule |
|---|---|
| ackSystemId | |
| ackUserId | |
| alarmPattern | |
| @type | |

#### Usage samples

Example of a request for creating a task to acknowledge alarms:

#### Request

```
POST /ackAlarm
Content-Type: application/json

{
    "ackSystemId": "ems-1",
    "ackUserId": "bob@example.net",
    "ackTime": "2019-07-03T03:33:12.623Z",
    "@type": "AckAlarm",
    "alarmPattern": [
        {
            "@type": "AlarmRef",
            "id": "8675309"
        },
        {
            "@type": "AlarmRef",
            "id": "8675399"
        }
    ]
}
```

#### Response

```
201

{
```

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```
    "id": "1038",
    "href": "/alarmManagement/v5/ackAlarm/1038",
    "@type": "AckAlarm",
    "ackSystemId": "ems-1",
    "ackUserId": "bob@example.net",
    "ackTime": "2019-07-03T03:33:12.623Z",
    "alarmPattern": [
        {
            "@type": "Alarm",
            "id": "8675309"
        },
        {
            "@type": "Alarm",
            "id": "8675399"
        }
    ],
    "state": "done",
    "ackedAlarm": [
        {
            "id": "8675309",
            "href": "/alarmManagement/v5/alarm/8675309",
            "@type": "Alarm",
            "externalAlarmId": "5551212",
            "state": "updated",
            "alarmType": "environmentalAlarm",
            "perceivedSeverity": "major",
            "probableCause": "rectifierLowVoltage",
            "specificProblem": "ps=3,sl=1,in=8",
            "alarmedObjectType": "Rectifier",
            "alarmedObject": {
                "@type": "Resource",
                "id": "93051825",
                "href": "/resourceInventoryManagement/v5/resource/93051825"
            },
            "sourceSystemId": "ems-1",
            "alarmDetails": "voltage=204",
            "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
            "alarmReportingTime": "2019-07-03T03:32:17.552Z",
            "alarmChangedTime": "2019-07-03T03:32:24.715Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackTime": "2019-07-03T03:33:12.623Z",
            "ackState": "acknowledged",
            "correlatedAlarm": [
                {
                    "@type": "Alarm",
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/868675399"
                }
            ]
        },
        {
            "id": "8675399",
            "href": "/alarmManagement/v5/alarm/8675399",
            "@type": "Alarm",
            "externalAlarmId": "5551234",
            "state": "raised",
            "alarmType": "Equipment Alarm",
            "perceivedSeverity": "minor",
            "probableCause": "powerSupplyFailure",
            "specificProblem": "ch=1,ps=3",
            "alarmedObjectType": "Chassis",
            "alarmedObject": {
```


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```
                "@type": "Resource",
                "id": "95830731",
                "href": "/resourceInventoryManagement/v5/resource/95830731"
            },
            "sourceSystemId": "ems-1",
            "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
            "alarmReportingTime": "2019-07-03T03:32:18.102Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackTime": "2019-07-03T03:33:12.623Z",
            "ackState": "acknowledged",
            "parentAlarm": [
                {
                    "@type": "Alarm",
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/868675309"
                }
            ]
        }
    ]
}
```


## Operations on UnAckAlarm

### Retrieves a UnAckAlarm by ID

**`GET /unAckAlarm/{id}?fields=…`**

#### Description

This operation retrieves a UnAckAlarm entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

#### Usage samples

Example of a request for retrieving an unacknowledge alarms task:

#### Request

```
GET /unAckAlarm/1052
Content-Type: application/json
```

#### Response


```
200

{
    "id": "1052",
    "@type": "UnAckAlarm",
    "href": "/alarmManagement/v5/unAckAlarm/1052",
    "ackSystemId": "ems-1",
    "ackUserId": "bob@example.net",
    "ackTime": "2019-07-03T03:38:06.890Z",
    "alarmPattern": [
```


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```
        {
            "@type": "Alarm",
            "id": "8675309"
        },
        {
            "@type": "Alarm",
            "id": "8675399"
        }
    ],
    "state": "inProgress",
    "unAckedAlarm": []
}
```

## List or find UnAckAlarm objects

### `GET /unAckAlarm?fields=…`

#### Description

This operation list UnAckAlarm entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

#### Usage samples

Example of a request for retrieving a list of unacknowledge alarms tasks:

#### Request

```
GET /unAckAlarm
Content-Type: application/json
```

#### Response

```
200

[
    {
        "@type": "UnAckAlarm",
        "id": "1052",
        "href": "/alarmManagement/v5/unAckAlarm/1052",
        "ackSystemId": "ems-1",
        "ackUserId": "bob@example.net",
        "ackTime": "2019-07-03T03:38:06.890Z",
        "alarmPattern": [
            {
                "@type": "Alarm",
                "id": "8675309"
            },
            {
                "@type": "Alarm",
                "id": "8675399"
            }
        ],
        "state": "inProgress",
```

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```
        "unAckedAlarm": []
    }
]
```

## Creates a UnAckAlarm

### `POST /unAckAlarm?fields=…`

#### Description

This operation creates a UnAckAlarm entity.

#### Mandatory Attributes

| Mandatory Attributes | Rule |
|---|---|
| ackSystemId | |
| ackUserId | |
| alarmPattern | |
| @type | |

#### Usage samples

Example of a request for creating an UnAckAlarm:

#### Request

```
POST /unAckAlarm
Content-Type: application/json

{
    "@type": "UnAckAlarm",
    "ackSystemId": "ems-1",
    "ackUserId": "bob@example.net",
    "ackTime": "2019-07-03T03:38:06.890Z",
    "alarmPattern": [
        {
            "@type": "AlarmRef",
            "id": "8675309"
        },
        {
            "@type": "AlarmRef",
            "id": "8675399"
        }
    ]
}
```

#### Response

```
201

{
    "@type": "UnAckAlarm",
```

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```
    "id": "1052",
    "href": "/alarmManagement/v5/unAckAlarm/1052",
    "ackSystemId": "ems-1",
    "ackUserId": "bob@example.net",
    "ackTime": "2019-07-03T03:38:06.890Z",
    "alarmPattern": [
        {
            "@type": "AlarmRef",
            "id": "8675309"
        },
        {
            "@type": "AlarmRef",
            "id": "8675399"
        }
    ],
    "state": "done",
    "unAckedAlarm": [
        {
            "id": "8675309",
            "href": "/alarmManagement/v5/alarm/8675309",
            "@type": "Alarm",
            "externalAlarmId": "5551212",
            "state": "updated",
            "alarmType": "environmentalAlarm",
            "perceivedSeverity": "major",
            "probableCause": "rectifierLowVoltage",
            "specificProblem": "ps=3,sl=1,in=8",
            "alarmedObjectType": "Rectifier",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "93051825",
                "href": "/resourceInventoryManagement/v5/resource/93051825"
            },
            "sourceSystemId": "ems-1",
            "alarmDetails": "voltage=95",
            "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
            "alarmReportingTime": "2019-07-03T03:32:17.552Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackState": "unacknowledged",
            "correlatedAlarm": [
                {
                    "@type": "AlarmRef",
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/868675399"
                }
            ],
            "comment": [
                {
                    "@type": "Comment",
                    "userId": "bob@example.net",
                    "systemId": "ems-1",
                    "time": "2019-07-03T03:37:33.827Z",
                    "comment": "Dispatched"
                }
            ]
        },
        {
            "id": "8675399",
            "href": "/alarmManagement/v5/alarm/8675399",
            "@type": "Alarm",
            "externalAlarmId": "5551234",
```


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```
            "state": "raised",
            "alarmType": "equipmentAlarm",
            "perceivedSeverity": "minor",
            "probableCause": "powerSupplyFailure",
            "specificProblem": "ch=1,ps=3",
            "alarmedObjectType": "Chassis",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "95830731",
                "href": "/resourceInventoryManagement/v5/resource/95830731"
            },
            "sourceSystemId": "ems-1",
            "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
            "alarmReportingTime": "2019-07-03T03:32:18.102Z",
            "alarmChangedTime": "2019-07-03T03:32:52.744Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackState": "unacknowledged",
            "parentAlarm": [
                {
                    "@type": "AlarmRef",
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/868675309"
                }
            ],
            "comment": [
                {
                    "@type": "Comment",
                    "userId": "bob@example.net",
                    "systemId": "ems-1",
                    "time": "2019-07-03T03:37:33.827Z",
                    "comment": "Dispatched"
                }
            ]
        }
    ]
}
```


## Operations on ClearAlarm

### List or find ClearAlarm objects

**`GET /clearAlarm?fields=…`**

#### Description

This operation list ClearAlarm entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

#### Usage samples

Example of a request for retrieving a list of clear alarms tasks:

#### Request

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```
GET /clearAlarm
Content-Type: application/json
```

### Response

```
200

[
    {
        "id": "1089",
        "href": "/alarmManagement/v5/clearAlarm/1089",
        "@type": "ClearAlarm",
        "clearSystemId": "ems-1",
        "clearUserId": "bob@example.net",
        "alarmClearedTime": "2019-07-03T03:55:45.937Z",
        "alarmPattern": [
            {
                "@type": "Alarm",
                "id": "8675309"
            },
            {
                "@type": "Alarm",
                "id": "8675399"
            }
        ],
        "state": "done",
        "clearedAlarm": [
            {
                "id": "8675309",
                "href": "/alarmManagement/v5/alarm/8675309",
                "@type": "Alarm",
                "externalAlarmId": "5551212",
                "state": "cleared",
                "alarmType": "environmentalAlarm",
                "perceivedSeverity": "cleared",
                "probableCause": "rectifierLowVoltage",
                "specificProblem": "ps=3,sl=1,in=8",
                "alarmedObjectType": "Rectifier",
                "alarmedObject": {
                    "@type": "AlarmedObjectRef",
                    "@referredType": "Resource",
                    "id": "93051825",
                    "href": "/resourceInventoryManagement/v5/resource/93051825"
                },
                "sourceSystemId": "ems-1",
                "alarmDetails": "voltage=95",
                "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
                "alarmReportingTime": "2019-07-03T03:32:17.552Z",
                "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                "ackSystemId": "ems-1",
                "ackUserId": "bob@example.net",
                "ackState": "unacknowledged",
                "comment": [
                    {
                        "@type": "Comment",
                        "userId": "bob@example.net",
                        "systemId": "ems-1",
                        "time": "2019-07-03T03:37:33.827Z",
                        "comment": "Dispatched"
```

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```
                    }
                ],
                "clearSystemId": "ems-1",
                "clearUserId": "bob@example.net",
                "alarmClearedTime": "2019-07-03T03:55:45.937Z"
            },
            {
                "id": "8675399",
                "href": "/alarmManagement/v5/alarm/8675399",
                "@type": "Alarm",
                "externalAlarmId": "5551234",
                "state": "cleared",
                "alarmType": "equipmentAlarm",
                "perceivedSeverity": "cleared",
                "probableCause": "powerSupplyFailure",
                "specificProblem": "ch=1,ps=3",
                "alarmedObjectType": "Chassis",
                "alarmedObject": {
                    "@type": "AlarmedObjectRef",
                    "@referredType": "Resource",
                    "id": "95830731",
                    "href": "/resourceInventoryManagement/v5/resource/95830731"
                },
                "sourceSystemId": "ems-1",
                "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
                "alarmReportingTime": "2019-07-03T03:32:18.102Z",
                "alarmChangedTime": "2019-07-03T03:42:12.861Z",
                "ackSystemId": "ems-1",
                "ackUserId": "bob@example.net",
                "ackState": "unacknowledged",
                "comment": [
                    {
                        "@type": "Comment",
                        "userId": "bob@example.net",
                        "systemId": "ems-1",
                        "time": "2019-07-03T03:37:33.827Z",
                        "comment": "Dispatched"
                    }
                ],
                "clearSystemId": "ems-1",
                "clearUserId": "bob@example.net",
                "alarmClearedTime": "2019-07-03T03:55:45.937Z"
            }
        ]
    }
]
```


## Retrieves a ClearAlarm by ID

**`GET /clearAlarm/{id}?fields=…`**

### Description

This operation retrieves a ClearAlarm entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

### Usage samples

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Example of a request for retrieving a clear alarms task:

### Request

```
GET /clearAlarm/1089
Content-Type: application/json
```

### Response

```
200

{
    "id": "1089",
    "href": "/alarmManagement/v5/clearAlarm/1089",
    "@type": "ClearAlarm",
    "clearSystemId": "ems-1",
    "clearUserId": "bob@example.net",
    "alarmClearedTime": "2019-07-03T03:55:45.937Z",
    "alarmPattern": [
        {
            "@type": "Alarm",
            "id": "8675309"
        },
        {
            "@type": "Alarm",
            "id": "8675399"
        }
    ],
    "state": "done",
    "clearedAlarm": [
        {
            "id": "8675309",
            "href": "/alarmManagement/v5/alarm/8675309",
            "@type": "Alarm",
            "externalAlarmId": "5551212",
            "state": "cleared",
            "alarmType": "environmentalAlarm",
            "perceivedSeverity": "cleared",
            "probableCause": "rectifierLowVoltage",
            "specificProblem": "ps=3,sl=1,in=8",
            "alarmedObjectType": "Rectifier",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "93051825",
                "href": "/resourceInventoryManagement/v5/resource/93051825"
            },
            "sourceSystemId": "ems-1",
            "alarmDetails": "voltage=95",
            "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
            "alarmReportingTime": "2019-07-03T03:32:17.552Z",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackState": "unacknowledged",
            "comment": [
                {
                    "@type": "Comment",
                    "userId": "bob@example.net",
```

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```
                    "systemId": "ems-1",
                    "time": "2019-07-03T03:37:33.827Z",
                    "comment": "Dispatched"
                }
            ],
            "clearSystemId": "ems-1",
            "clearUserId": "bob@example.net",
            "alarmClearedTime": "2019-07-03T03:55:45.937Z"
        },
        {
            "id": "8675399",
            "href": "/alarmManagement/v5/alarm/8675399",
            "@type": "Alarm",
            "externalAlarmId": "5551234",
            "state": "cleared",
            "alarmType": "equipmentAlarm",
            "perceivedSeverity": "cleared",
            "probableCause": "powerSupplyFailure",
            "specificProblem": "ch=1,ps=3",
            "alarmedObjectType": "Chassis",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "95830731",
                "href": "/resourceInventoryManagement/v5/resource/95830731"
            },
            "sourceSystemId": "ems-1",
            "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
            "alarmReportingTime": "2019-07-03T03:32:18.102Z",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackState": "unacknowledged",
            "comment": [
                {
                    "@type": "Comment",
                    "userId": "bob@example.net",
                    "systemId": "ems-1",
                    "time": "2019-07-03T03:37:33.827Z",
                    "comment": "Dispatched"
                }
            ],
            "clearSystemId": "ems-1",
            "clearUserId": "bob@example.net",
            "alarmClearedTime": "2019-07-03T03:55:45.937Z"
        }
    ]
}
```


## Creates a ClearAlarm

**`POST /clearAlarm?fields=…`**

### Description

This operation creates a ClearAlarm entity.

### Mandatory Attributes

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| Mandatory Attributes | Rule |
|---|---|
| alarmClearedTime | |
| alarmPattern | |
| clearSystemId | |
| clearUserId | |
| @type | |

### Usage samples

Example of a request for creating an ClearAlarm:

### Request

```
POST /clearAlarm
Content-Type: application/json

{
    "@type": "ClearAlarm",
    "clearSystemId": "ems-1",
    "clearUserId": "bob@example.net",
    "alarmClearedTime": "2019-07-03T03:55:45.937Z",
    "alarmPattern": [
        {
            "@type": "AlarmRef",
            "id": "8675309"
        },
        {
            "@type": "AlarmRef",
            "id": "8675399"
        }
    ]
}
```

### Response

```
201

{
    "id": "1089",
    "href": "/alarmManagement/v5/clearAlarm/1089",
    "@type": "ClearAlarm",
    "clearSystemId": "ems-1",
    "clearUserId": "bob@example.net",
    "alarmClearedTime": "2019-07-03T03:55:45.937Z",
    "alarmPattern": [
        {
            "@type": "AlarmRef",
            "id": "8675309"
        },
        {
            "@type": "AlarmRef",
            "id": "8675399"
        }
```

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```
    ],
    "state": "done",
    "clearedAlarm": [
        {
            "id": "8675309",
            "href": "/alarmManagement/v5/alarm/8675309",
            "@type": "Alarm",
            "externalAlarmId": "5551212",
            "state": "cleared",
            "alarmType": "environmentalAlarm",
            "perceivedSeverity": "cleared",
            "probableCause": "rectifierLowVoltage",
            "specificProblem": "ps=3,sl=1,in=8",
            "alarmedObjectType": "Rectifier",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "93051825",
                "href": "/resourceInventoryManagement/v5/resource/93051825"
            },
            "sourceSystemId": "ems-1",
            "alarmDetails": "voltage=95",
            "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
            "alarmReportingTime": "2019-07-03T03:32:17.552Z",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackState": "unacknowledged",
            "comment": [
                {
                    "@type": "Comment",
                    "userId": "bob@example.net",
                    "systemId": "ems-1",
                    "time": "2019-07-03T03:37:33.827Z",
                    "comment": "Dispatched"
                }
            ],
            "clearSystemId": "ems-1",
            "clearUserId": "bob@example.net",
            "alarmClearedTime": "2019-07-03T03:55:45.937Z"
        },
        {
            "id": "8675399",
            "href": "/alarmManagement/v5/alarm/8675399",
            "@type": "Alarm",
            "externalAlarmId": "5551234",
            "state": "cleared",
            "alarmType": "equipmentAlarm",
            "perceivedSeverity": "cleared",
            "probableCause": "powerSupplyFailure",
            "specificProblem": "ch=1,ps=3",
            "alarmedObjectType": "Chassis",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "95830731",
                "href": "/resourceInventoryManagement/v5/resource/95830731"
            },
            "sourceSystemId": "ems-1",
            "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
            "alarmReportingTime": "2019-07-03T03:32:18.102Z",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "ackSystemId": "ems-1",
```


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```
            "ackUserId": "bob@example.net",
            "ackState": "unacknowledged",
            "comment": [
                {
                    "@type": "Comment",
                    "userId": "bob@example.net",
                    "systemId": "ems-1",
                    "time": "2019-07-03T03:37:33.827Z",
                    "comment": "Dispatched"
                }
            ],
            "clearSystemId": "ems-1",
            "clearUserId": "bob@example.net",
            "alarmClearedTime": "2019-07-03T03:55:45.937Z"
        }
    ]
}
```

## Operations on CommentAlarm

### List or find CommentAlarm objects

**`GET /commentAlarm?fields=…`**

#### Description

This operation list CommentAlarm entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

#### Usage samples

Example of a request for retrieving a list of comment alarms tasks:

#### Request

```
GET /commentAlarm
Content-Type: application/json
```

#### Response

```
200

[
    {
        "@type": "CommentAlarm",
        "id": "1045",
        "href": "/alarmManagement/v5/ackAlarm/1045",
        "comment": {
            "@type": "Comment",
            "userId": "bob@example.net",
            "systemId": "ems-1",
            "time": "2019-07-03T03:37:33.827Z",
            "comment": "Dispatched"
        },
```

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```
        "alarmPattern": [
            {
                "@type": "Alarm",
                "id": "8675309"
            },
            {
                "@type": "Alarm",
                "id": "8675399"
            }
        ],
        "state": "inProgress",
        "commentedAlarm": []
    }
]
```

## Retrieves a CommentAlarm by ID

**`GET /commentAlarm/{id}?fields=…`**

### Description

This operation retrieves a CommentAlarm entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

### Usage samples

Example of a request for retrieving a comment alarms task:

### Request

```
GET /commentAlarm/1045
Content-Type: application/json
```

### Response

```
200

{
    "@type": "CommentAlarm",
    "id": "1045",
    "href": "/alarmManagement/v5/ackAlarm/1045",
    "comment": {
        "@type": "Comment",
        "userId": "bob@example.net",
        "systemId": "ems-1",
        "time": "2019-07-03T03:37:33.827Z",
        "comment": "Dispatched"
    },
    "alarmPattern": [
        {
            "@type": "Alarm",
            "id": "8675309"
        },
        {
```

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```
            "@type": "Alarm",
            "id": "8675399"
        }
    ],
    "state": "inProgress",
    "commentedAlarm": []
}
```

## Creates a CommentAlarm

**`POST /commentAlarm?fields=…`**

### Description

This operation creates a CommentAlarm entity.

### Mandatory Attributes

| Mandatory Attributes | Rule |
|---|---|
| alarmPattern | |
| comment | |
| comment.@type | |
| @type | |

### Usage samples

Example of a request for creating an CommentAlarm:

### Request

```
POST /commentAlarm
Content-Type: application/json

{
    "@type": "CommentAlarm",
    "comment": {
        "@type": "Comment",
        "userId": "bob@example.net",
        "systemId": "ems-1",
        "time": "2019-07-03T03:37:33.827Z",
        "comment": "Dispatched"
    },
    "alarmPattern": [
        {
            "@type": "Alarm",
            "id": "8675309"
        },
        {
            "@type": "Alarm",
            "id": "8675399"
        }
    ]
}
```

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### Response

```
201

{
    "@type": "CommentAlarm",
    "id": "1045",
    "href": "/alarmManagement/v5/ackAlarm/1045",
    "comment": {
        "@type": "Comment",
        "userId": "bob@example.net",
        "systemId": "ems-1",
        "time": "2019-07-03T03:37:33.827Z",
        "comment": "Dispatched"
    },
    "alarmPattern": [
        {
            "@type": "Alarm",
            "id": "8675309"
        },
        {
            "@type": "Alarm",
            "id": "8675399"
        }
    ],
    "state": "inProgress",
    "commentedAlarm": []
}
```

## Operations on GroupAlarm

### List or find GroupAlarm objects

**`GET /groupAlarm?fields=…`**

#### Description

This operation list GroupAlarm entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

#### Usage samples

Example of a request for retrieving a list of group alarms tasks:

#### Request

```
GET /groupAlarm
Content-Type: application/json
```

#### Response

```
200
```

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```
[
    {
        "id": "1029",
        "@type": "GroupAlarm",
        "href": "/alarmManagement/v5/groupAlarm/1029",
        "sourceSystemId": "ems-1",
        "alarmChangedTime": "2019-07-03T03:32:52.744Z",
        "parentAlarm": {
            "@type": "Alarm",
            "id": "8675309"
        },
        "correlatedAlarm": [
            {
                "@type": "Alarm",
                "id": "8675399"
            }
        ],
        "state": "inProgress",
        "groupedAlarm": []
    }
]
```


## Retrieves a GroupAlarm by ID

**`GET /groupAlarm/{id}?fields=…`**

### Description

This operation retrieves a GroupAlarm entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

### Usage samples

Example of a request for retrieving an group alarms task:

### Request

```
GET /groupAlarm/1029
Content-Type: application/json
```

### Response


```
200

{
    "id": "1029",
    "@type": "GroupAlarm",
    "href": "/alarmManagement/v5/groupAlarm/1029",
    "sourceSystemId": "ems-1",
    "alarmChangedTime": "2019-07-03T03:32:52.744Z",
    "parentAlarm": {
        "@type": "Alarm",
        "id": "8675309"
```


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```
    },
    "correlatedAlarm": [
        {
            "@type": "Alarm",
            "id": "8675399"
        }
    ],
    "state": "inProgress",
    "groupedAlarm": []
}
```

## Creates a GroupAlarm

### `POST /groupAlarm?fields=…`

#### Description

This operation creates a GroupAlarm entity.

#### Mandatory Attributes

| Mandatory Attributes | Rule |
|---|---|
| alarmChangedTime | |
| correlatedAlarm | |
| parentAlarm | |
| sourceSystemId | |
| @type | |

#### Usage samples

Example of a request for creating a group of alarms:

#### Request

```
POST /groupAlarm
Content-Type: application/json

{
    "@type": "GroupAlarm",
    "sourceSystemId": "ems-1",
    "alarmChangedTime": "2019-07-03T03:32:52.744Z",
    "parentAlarm": {
        "@type": "AlarmRef",
        "id": "8675309"
    },
    "correlatedAlarm": [
        {
            "@type": "AlarmRef",
            "id": "8675399"
        }
    ]
}
```

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## Response

```
201

{
    "id": "1029",
    "@type": "GroupAlarm",
    "href": "/alarmManagement/v5/groupAlarm/1029",
    "sourceSystemId": "ems-1",
    "alarmChangedTime": "2019-07-03T03:32:52.744Z",
    "parentAlarm": {
        "@type": "AlarmRef",
        "id": "8675309"
    },
    "correlatedAlarm": [
        {
            "@type": "AlarmRef",
            "id": "8675399"
        }
    ],
    "state": "done",
    "groupedAlarm": [
        {
            "id": "8675309",
            "href": "/alarmManagement/v5/alarm/8675309",
            "@type": "Alarm",
            "externalAlarmId": "5551212",
            "state": "updated",
            "alarmType": "environmentalAlarm",
            "perceivedSeverity": "major",
            "probableCause": "rectifierLowVoltage",
            "specificProblem": "ps=3,sl=1,in=8",
            "alarmedObjectType": "Rectifier",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "93051825",
                "href": "/resourceInventoryManagement/v5/resource/93051825"
            },
            "sourceSystemId": "ems-1",
            "alarmDetails": "voltage=95",
            "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
            "alarmReportingTime": "2019-07-03T03:32:17.552Z",
            "alarmChangedTime": "2019-07-03T03:32:52.744Z",
            "correlatedAlarm": [
                {
                    "@type": "AlarmRef",
                    "id": "8675399",
                    "href": "/alarmManagement/v5/alarm/868675399"
                }
            ]
        },
        {
            "id": "8675399",
            "href": "/alarmManagement/v5/alarm/8675399",
            "@type": "Alarm",
            "externalAlarmId": "5551234",
            "state": "raised",
            "alarmType": "equipmentAlarm",
            "perceivedSeverity": "minor",
            "probableCause": "powerSupplyFailure",
```

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```
            "specificProblem": "ch=1,ps=3",
            "alarmedObjectType": "Chassis",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "95830731",
                "href": "/resourceInventoryManagement/v5/resource/95830731"
            },
            "sourceSystemId": "ems-1",
            "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
            "alarmReportingTime": "2019-07-03T03:32:18.102Z",
            "alarmChangedTime": "2019-07-03T03:32:52.744Z",
            "parentAlarm": [
                {
                    "@type": "AlarmRef",
                    "id": "8675309",
                    "href": "/alarmManagement/v5/alarm/868675309"
                }
            ]
        }
    ]
}
```


## Operations on UnGroupAlarm

### List or find UnGroupAlarm objects

**`GET /unGroupAlarm?fields=…`**

#### Description

This operation list UnGroupAlarm entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

#### Usage samples

Example of a request for retrieving a list of unhgroup alarms tasks:

#### Request

```
GET /unGroupAlarm
Content-Type: application/json
```

#### Response


```
200

[
    {
        "id": "1069",
        "@type": "UnGroupAlarm",
        "href": "/alarmManagement/v5/unGroupAlarm/1069",
        "sourceSystemId": "ems-1",
        "alarmChangedTime": "2019-07-03T03:42:12.861Z",
```


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```
        "parentAlarm": {
            "@type": "Alarm",
            "id": "8675309"
        },
        "correlatedAlarm": [
            {
                "@type": "Alarm",
                "id": "8675399"
            },
            {
                "@type": "Alarm",
                "id": "8675399"
            }
        ],
        "state": "inProgress",
        "unGroupedAlarm": []
    }
]
```

## Retrieves a UnGroupAlarm by ID

**`GET /unGroupAlarm/{id}?fields=…`**

### Description

This operation retrieves a UnGroupAlarm entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

### Usage samples

Example of a request for retrieving an ungroup alarms task:

### Request

```
GET /unGroupAlarm/1069
Content-Type: application/json
```

### Response

```
200

{
    "id": "1069",
    "@type": "UnGroupAlarm",
    "href": "https://host:port/alarmManagement/v5/groupAlarm/1069",
    "sourceSystemId": "ems-1",
    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
    "parentAlarm": {
        "@type": "Alarm",
        "id": "8675309"
    },
    "correlatedAlarm": [
        {
            "@type": "Alarm",
```

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```
            "id": "8675399"
        },
        {
            "@type": "Alarm",
            "id": "8675399"
        }
    ],
    "state": "inProgress",
    "unGroupedAlarm": []
}
```

## Creates a UnGroupAlarm

### POST /unGroupAlarm?fields=…

#### Description

This operation creates a UnGroupAlarm entity.

#### Mandatory Attributes

| Mandatory Attributes | Rule |
|---|---|
| alarmChangedTime | |
| correlatedAlarm | |
| parentAlarm | |
| sourceSystemId | |
| @type | |

#### Usage samples

Example of a request for creating an UnGroupAlarm:

#### Request

```
POST /unGroupAlarm
Content-Type: application/json

{
    "@type": "UnGroupAlarm",
    "sourceSystemId": "ems-1",
    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
    "parentAlarm": {
        "@type": "Alarm",
        "id": "8675309"
    },
    "correlatedAlarm": [
        {
            "@type": "Alarm",
            "id": "8675399"
        },
        {
            "@type": "Alarm",
            "id": "8675399"
```

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```
        }
    ]
}
```

### Response

```
201

{
    "id": "1069",
    "@type": "UnGroupAlarm",
    "href": "/alarmManagement/v5/unGroupAlarm/1069",
    "sourceSystemId": "ems-1",
    "alarmChangedTime": "2019-07-03T03:42:12.861Z",
    "parentAlarm": {
        "@type": "Alarm",
        "id": "8675309"
    },
    "correlatedAlarm": [
        {
            "@type": "Alarm",
            "id": "8675399"
        },
        {
            "@type": "Alarm",
            "id": "8675399"
        }
    ],
    "state": "done",
    "unGroupedAlarm": [
        {
            "id": "8675309",
            "href": "/alarmManagement/v5/alarm/8675309",
            "@type": "Alarm",
            "externalAlarmId": "5551212",
            "state": "updated",
            "alarmType": "environmentalAlarm",
            "perceivedSeverity": "major",
            "probableCause": "rectifierLowVoltage",
            "specificProblem": "ps=3,sl=1,in=8",
            "alarmedObjectType": "Rectifier",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "93051825",
                "href": "/resourceInventoryManagement/v5/resource/93051825"
            },
            "sourceSystemId": "ems-1",
            "alarmDetails": "voltage=95",
            "alarmRaisedTime": "2019-07-03T03:32:17.235Z",
            "alarmReportingTime": "2019-07-03T03:32:17.552Z",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackState": "unacknowledged",
            "comment": [
                {
                    "@type": "Comment",
                    "userId": "bob@example.net",
                    "systemId": "ems-1",
```

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```
                    "time": "2019-07-03T03:37:33.827Z",
                    "comment": "Dispatched"
                }
            ]
        },
        {
            "id": "8675399",
            "href": "/alarmManagement/v5/alarm/8675399",
            "@type": "Alarm",
            "externalAlarmId": "5551234",
            "state": "updated",
            "alarmType": "equipmentAlarm",
            "perceivedSeverity": "minor",
            "probableCause": "powerSupplyFailure",
            "specificProblem": "ch=1,ps=3",
            "alarmedObjectType": "Chassis",
            "alarmedObject": {
                "@type": "AlarmedObjectRef",
                "@referredType": "Resource",
                "id": "95830731",
                "href": "/resourceInventoryManagement/v5/resource/95830731"
            },
            "sourceSystemId": "ems-1",
            "alarmRaisedTime": "2019-07-03T03:32:17.843Z",
            "alarmReportingTime": "2019-07-03T03:32:18.102Z",
            "alarmChangedTime": "2019-07-03T03:42:12.861Z",
            "ackSystemId": "ems-1",
            "ackUserId": "bob@example.net",
            "ackState": "unacknowledged",
            "comment": [
                {
                    "@type": "Comment",
                    "userId": "bob@example.net",
                    "systemId": "ems-1",
                    "time": "2019-07-03T03:37:33.827Z",
                    "comment": "Dispatched"
                }
            ]
        }
    ]
}
```


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# API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

## Register listener

### POST /hub

### Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

### Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

### Usage Samples

Here's an example of a request for registering a listener.

### Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

### Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

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## Unregister listener

### DELETE /hub/{id}

### Description

Clears the communication endpoint address that was set by creating the Hub..

### Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

### Usage Samples

Here's an example of a request for un-registering a listener.

### Request

```
DELETE /api/hub/42

Accept: application/json
```

### Response

```
204
```

## Publish Event to listener

### POST /client/listener

### Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

### Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

### Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

### Request

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```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

### Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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# Acknowledgements

## Release History

| Release Number | Date | Release led by: | Description |
|---|---|---|---|
| Release 17.0.0 | 20-Sep-2017 | Yuval Stein, TEOCO<br><br>Pierre Gauthier, TM Forum <pgauthier@tmforum.org> | First Release of the Document. |
| Release 17.0.1 | 04-Dec-2017 | Adrienne Walcott,TM Forum | Updated to reflect TM Forum Approved Status |
| | 14-Nov-2019 | Vance Shipley, SigScale <vances@sigscale.com> | Schemafication project. |
| Pre-production | 03-Apr-2020 | Vance Shipley, SigScale <vances@sigscale.com> | Update for User Guide format. |
| Production | 25-May-2020 | Adrienne Walcott | Updated to reflect TM Forum Approved Status |
| Production v5 | 19-Sep-2023 | Pierre Gauthier, TM Forum <pgauthier@tmforum.org> | |

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